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# Reliability Conditions for Interpreting Daily NASA Black Marble Nighttime Lights in Cloud-Impacted Tropical Regions



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# Supplementary Materials: Reliability Conditions for Interpreting Daily NASA Black Marble Nighttime Lights in Cloud-Impacted Tropical Regions

**Authors: Rene Jr. L. Principe, Karin Reinke, and Simon Jones**

This document provides supporting figures, tables, and diagnostic analyses that complement the main manuscript. It expands key results by presenting full-resolution metrics, illustrative examples, and methodological details that reinforce the reliability framework.

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## I. Points of Interest and Spatial Supports

To complement settlement-scale diagnostics, ten representative points-of-interest (POIs) were defined across Samar–Leyte to characterise local-scale observability under varying settlement intensity and land-cover conditions. These POIs provide controlled reference locations for examining radiance behaviour at fine spatial scales, independent of thematic aggregation.

The selected sites span three settlement contexts. Urban POIs include Tacloban City Centre, Ormoc City Centre, and Baybay City Centre, representing high-density electrified environments. Municipal and peri-urban POIs include Palo, Alang-Alang, and Guiuan, capturing intermediate radiance levels and heterogeneous infrastructure. Non-settlement reference sites include Samar Forest Reserve, Mt. Nacolod Upland, Central Leyte Forest, and Southern Leyte Gulf, representing minimally illuminated or uninhabited conditions.

All POIs were aligned to the native 500 m VIIRS grid to ensure pixel-consistent extraction from the NASA Black Marble VNP46A2 product (Román et al., 2018). Radiance was summarised



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using three nested spatial supports: 1×1 (single pixel), 3×3 (9 pixels), and 5×5 (25 pixels). This nested design isolates the effect of spatial aggregation on signal stability. The 1×1 support preserves strict locality but is highly sensitive to cloud masking and transient variability. Larger supports increase spatial support and reduce local radiance variability through averaging, but do not consistently increase completeness.

These POI-based diagnostics are used exclusively to examine fine-scale observability and kernel effects; settlement-level ranking and alignment are performed using GHSL-stratified aggregation in the main manuscript.

Table S1. Points of interest (POIs) used for pixel- and patch-level observability diagnostics.

| Group | POI name | Coordinates | Rationale |
|---|---|---|---|
| City Centres | Tacloban City Centre | [125.0015, 11.2434] | Regional capital; severely affected by Haiyan; dense urban lighting |
| | Ormoc City Centre | [124.6068, 11.0088] | West Leyte urban hub; port; stable electrification |
| | Baybay City Centre | [124.7989, 10.6766] | Smaller coastal city; consistent urban lights |
| Municipal Centres | Palo | [124.9904, 11.1577] | Suburban east of Tacloban; major resettlement areas |
| | Alang-Alang | [124.8465, 11.2071] | Inland peri-urban near Tacloban |
| | Guiuan | [125.7232, 11.0312] | Southern Samar town; near Haiyan landfall |
| Non-settlement Areas | Samar Forest Reserve | [125.1870, 11.7740] | Remote forest; negligible light |
| | Mt. Nacolod Upland | [125.3100, 11.4960] | Rugged uplands; uninhabited |
| | Central Leyte Forest | [124.8710, 10.8180] | Forest patch; minimal infrastructure |
| | Southern Leyte Gulf | [125.3222, 10.8153] | Offshore dark reference; no settlements |

# II. Anatomy of NTL Observability and Gap-Filling Effects

This section presents diagnostic examples illustrating how observational completeness and temporal interpolation jointly shape daily NTL time series across spatial scales. The DNB-BRDF and gap-filled layers form part of the daily NASA Black Marble VNP46A2 product used in this analysis (Román et al., 2018). The figures distinguish between regionally aggregated behaviour and pixel-level dynamics. Figure S1 shows mean DNB–BRDF and gap-filled radiance aggregated over the entire Samar–Leyte domain, alongside spatial completeness. At this regional scale, the observed series captures the abrupt radiance decline associated with Typhoon Haiyan, but exhibits strong temporal irregularity driven by fluctuating valid-pixel availability.



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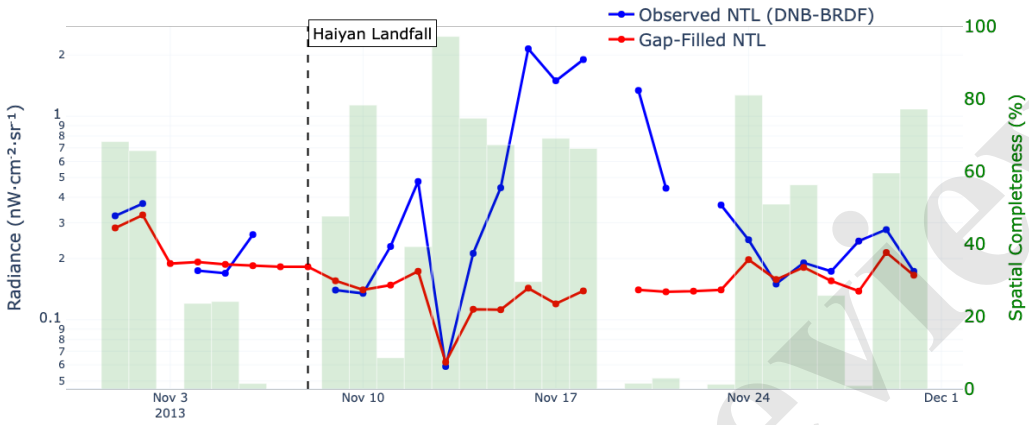


***Figure S1.*** *Regional-scale mean DNB–BRDF (blue) and gap-filled radiance (red) for Samar–Leyte, with spatial completeness (%) shown as background bars. The dashed line marks Typhoon Haiyan landfall.*

The gap-filled series, by contrast, produces a smoother and more continuous trajectory with reduced short-term variability. Periods of largest divergence between the two series align with low spatial completeness, indicating that interpolation effects are strongest when observational support is weakest. This demonstrates how gap-filling can preserve continuity at the expense of attenuating event-driven signal magnitude.

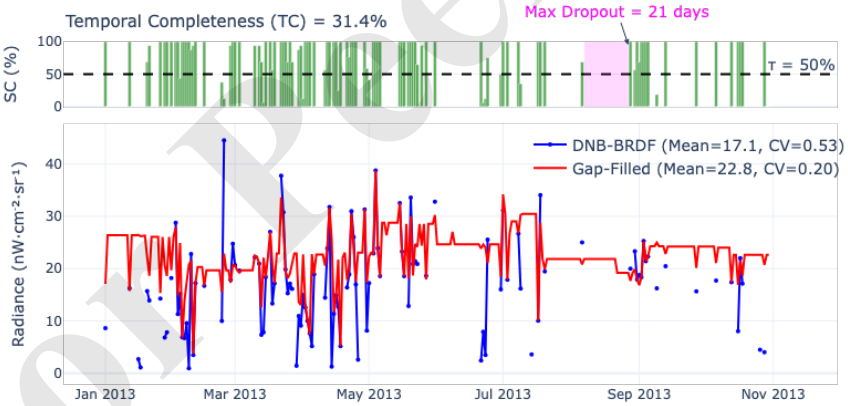


***Figure S2.*** *Pixel-scale observability diagnostics for a 1×1 extraction at Tacloban City centre. Top: temporal completeness relative to a spatial threshold, showing the maximum uninterrupted observation gap. Bottom: observed and gap-filled radiance time series with corresponding variability statistics.*

Figure S2 provides a pixel-scale perspective using a 1×1 extraction at the Tacloban City centre. The upper panel shows temporal completeness relative to a spatial threshold, highlighting



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intermittent sampling and extended dropout periods. The lower panel compares observed and gap-filled radiance at the same location. The observed series exhibits high variability and episodic excursions, while the gap-filled series suppresses these fluctuations and increases apparent stability. Differences in mean radiance and coefficient of variation reflect the smoothing imposed by temporal propagation. Importantly, dropout windows coincide with periods where gap-filled values dominate the apparent signal, illustrating how interpolation can reshape both variance and temporal structure at fine spatial supports.

Together, these diagnostics establish two key properties of daily NTL in cloud-affected environments. First, observational availability is temporally fragmented across both regional and pixel scales, with dropout durations that can overlap short-lived disruption and early recovery periods. Second, gap filling estimates unavailable values from surrounding spatial and temporal information but does not provide additional direct observations. More recent reconstruction methods likewise seek to produce continuous daily NTL sequences by exploiting spatiotemporal dependencies (Hao et al., 2023; Tan and Zhu, 2023). These behaviours reinforce the observability metrics and admissibility thresholds evaluated in subsequent sections.

## III. Long-Term Structure of Spatial Completeness

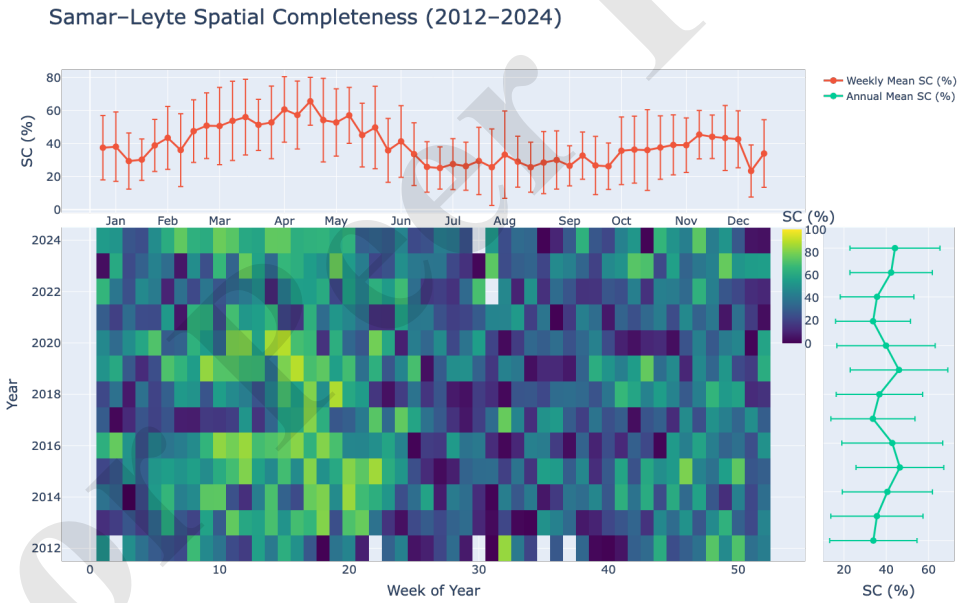


***Figure S3***. *Spatial completeness (%) for Samar–Leyte from 2012 to 2024. Top: weekly mean completeness with variability. Bottom-left: week-of-year heatmap showing seasonal and interannual structure. Bottom-right: annual mean completeness with variability ranges.*

To contextualise the event-scale diagnostics, Figure S3 characterises the temporal structure of spatial completeness across the full 2012–2024 NASA Black Marble VNP46A2 record for



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Samar–Leyte. Weekly mean completeness exhibits pronounced seasonal variability, with higher values typically observed during boreal spring and lower completeness during monsoonal periods. This seasonality is consistent across years, indicating that observational availability is systematically modulated by regional cloud climatology rather than episodic disturbances alone.

The heatmap representation reveals substantial interannual variability superimposed on this seasonal cycle. Periods of persistently low completeness occur across multiple consecutive weeks, while high-completeness intervals are comparatively limited. These patterns confirm that extended dropout windows identified at the event and pixel scales are embedded within a broader regime of seasonally structured observation loss.

Annual summaries further demonstrate that mean completeness remains moderate across all years, with substantial variability around the mean. The persistence of this variability over more than a decade indicates that incomplete sampling is not an anomaly associated with specific events but a stable characteristic of NTL observations in humid tropical environments.

These results reinforce that observational intermittency arises from large-scale atmospheric dynamics and operates consistently across temporal scales. Consequently, any reliability framework must account for both short-term dropout behaviour and long-term completeness structure.

## IV. Pre–Post Event Radiance Differences Across Spatial Supports

To quantify event-scale signal sensitivity, mean radiance differences (ΔNTL) before and after the 8 November 2013 landfall of Typhoon Haiyan were computed for urban centres, municipal centres, and non-settlement reference areas across three spatial supports (National Disaster Risk Reduction and Management Council, 2014). Supplementary Figure S4 summarises these pre–post contrasts.

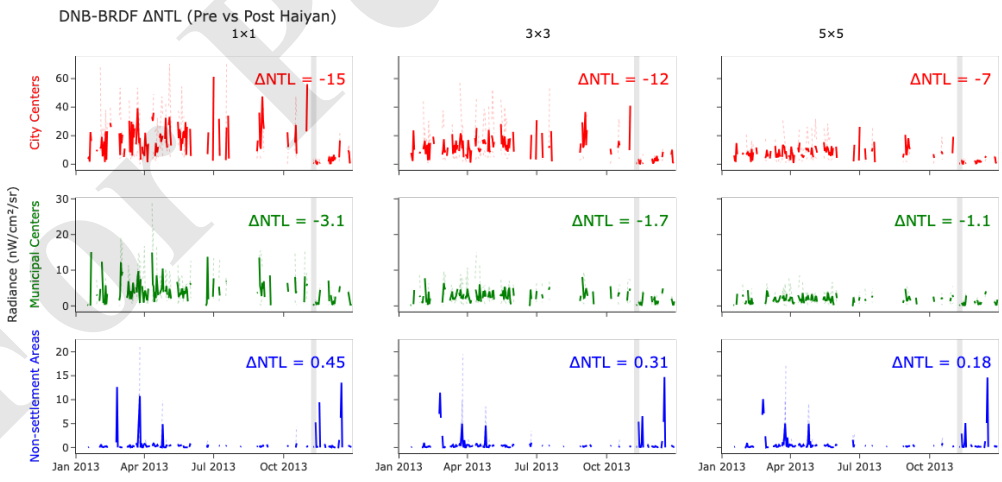




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***Figure S4.*** *Pre–post Typhoon Haiyan ΔNTL values for representative POIs across 1×1, 3×3, and 5×5 supports.*

ΔNTL was calculated as the mean post-event radiance (9 November 2013) minus the mean pre-event radiance (7 November 2013) using the reference landfall date of 8 November 2013, and retaining only dates satisfying SC ≥ τ.

Urban centres exhibit the largest negative ΔNTL values, indicating substantial radiance decline following the event. Municipal centres show moderate decreases, while non-settlement areas display near-zero or slightly positive changes, reflecting background noise and transient detections rather than infrastructure-driven signals.

Spatial support strongly influences magnitude estimates. Single-pixel (1×1) ΔNTL values show higher variance, whereas 5×5 supports produce more stable estimates but with slightly reduced effect magnitudes. This behaviour reflects a precision–stability trade-off inherent in spatial aggregation, where increased support reduces variability but smooths localized extremes.

These results reinforce the importance of spatial support selection in event-based NTL analysis and support the use of settlement-stratified aggregation to balance signal stability and sensitivity. Having established the influence of spatial support, we next examine the effect of temporal interpolation through gap-filled radiance.

## V. Kernel-Scale Comparison of DNB-BRDF and Gap-Filled Radiance

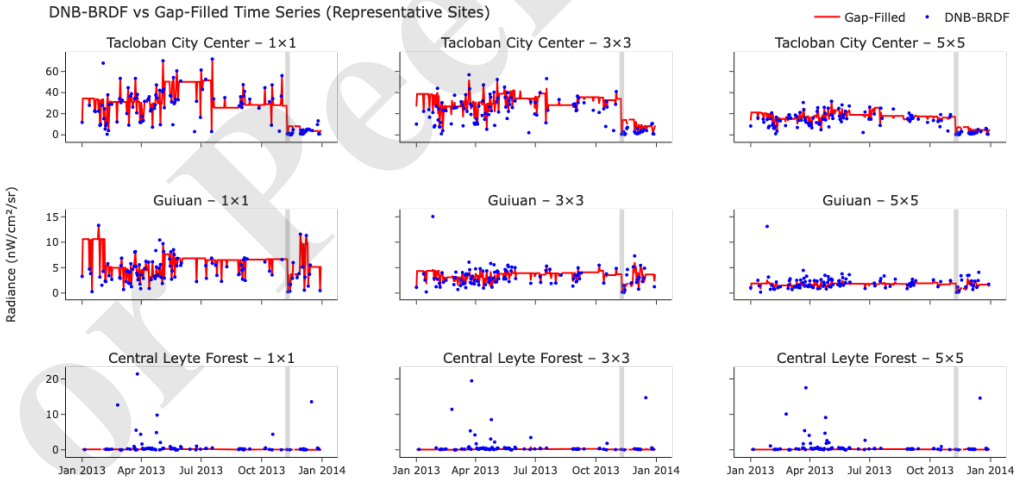


***Figure S5.*** *DNB-BRDF (blue) and gap-filled (red) radiance time series for representative POIs in 2013 across 1×1, 3×3, and 5×5 supports. The vertical grey bar indicates Typhoon Haiyan landfall.*

To examine the effect of temporal interpolation at fine spatial scales, DNB-BRDF and gap-filled radiance series were compared across representative POIs and spatial supports (1×1, 3×3, and



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5×5 pixels). Supplementary Figure S5 presents time series for an urban centre (Tacloban City), a municipal centre (Guiuan), and a non-settlement reference (Central Leyte Forest) during 2013.

Across all sites, the gap-filled series exhibits reduced short-term variability relative to DNB-BRDF, with the strongest smoothing observed at the 1×1 support. In urban locations, interpolation dampens the abrupt radiance decline immediately following Typhoon Haiyan, while in non-settlement areas it suppresses episodic spikes present in the raw DNB-BRDF signal. These effects indicate that gap-filling modifies both the magnitude and temporal sharpness of observed radiance dynamics.

Spatial aggregation (3×3 and 5×5) reduces noise in both series; however, structural differences between DNB-BRDF and gap-filled radiance persist across all supports. While aggregation improves stability, it does not reconcile discrepancies introduced by temporal imputation.

These results show that gap-filling increases apparent temporal continuity but can attenuate abrupt change signals that are critical for outage detection, particularly at finer spatial scales.

# VI. Full POI-Level Observability Metrics

Table S2 reports spatial completeness (SC), temporal completeness (TC), and maximum consecutive dropout duration for all ten POIs across three spatial supports (1×1, 3×3, and 5×5 pixels), derived from quality-screened DNB-BRDF observations.

***Table S2****. Spatial completeness (%), temporal completeness (%), and maximum consecutive dropout duration (days) for all POIs across 1×1, 3×3, and 5×5 supports.*

| | **Spatial Completeness (%)** | | | **Temporal Completeness (%)** | | | **Maximum Dropout Duration (days)** | | |
|---|---|---|---|---|---|---|---|---|---|
| | 1×1 | 3×3 | 5×5 | 1×1 | 3×3 | 5×5 | 1×1 | 3×3 | 5×5 |
| **City Centres (AVG)** | **31.1** | **28.9** | **26.5** | **31.1** | **30.3** | **28.9** | **14** | **14** | **14** |
| Tacloban City Centre | 32.6 | 30.8 | 28.7 | 32.6 | 31.8 | 29.9 | 11 | 11 | 13 |
| Ormoc City Centre | 32.3 | 29.4 | 27.0 | 32.3 | 32.1 | 31.0 | 15 | 15 | 15 |
| Baybay City Centre | 28.5 | 26.5 | 23.8 | 28.5 | 27.1 | 26.0 | 15 | 15 | 15 |
| **Municipal Centres (AVG)** | **33.5** | **31.9** | **30.2** | **33.5** | **33.1** | **32.7** | **13** | **13** | **13** |
| Palo | 34.8 | 33.7 | 32.8 | 34.8 | 34.0 | 34.0 | 13 | 13 | 13 |
| Alang Alang | 35.3 | 33.6 | 32.7 | 35.3 | 35.6 | 34.5 | 11 | 11 | 11 |
| Guiuan | 30.4 | 28.3 | 24.9 | 30.4 | 29.6 | 29.6 | 15 | 15 | 15 |
| **Non-settlement Areas (AVG)** | **26.7** | **24.9** | **24.2** | **26.7** | **26.4** | **26.2** | **27** | **26** | **27** |
| Samar Forest Reserve | 34.5 | 33.6 | 33.0 | 34.5 | 35.3 | 34.5 | 12 | 12 | 15 |
| Mt. Nacolod Upland | 35.6 | 32.8 | 32.0 | 35.6 | 35.1 | 35.3 | 16 | 13 | 13 |
| Central Leyte Forest | 23.6 | 21.8 | 21.1 | 23.6 | 23.3 | 23.3 | 18 | 18 | 18 |
| Southern Leyte Gulf | 13.2 | 11.4 | 10.9 | 13.2 | 11.8 | 11.5 | 61 | 62 | 62 |

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Across POIs, completeness metrics show relatively narrow ranges, but maximum dropout duration varies more substantially. Locations with similar SC and TC values may exhibit markedly different maximum gap lengths, indicating that completeness alone does not fully capture temporal continuity. Southern Leyte Gulf exhibits both the lowest completeness and the longest dropout duration among the reference sites.

Increasing spatial support produces only small changes in completeness, which generally declines slightly from the 1×1 to 5×5 grouped summaries, while maximum dropout duration remains largely unchanged. This indicates that increasing kernel support modifies the spatial averaging of retained observations but does not increase overall completeness or resolve extended observation gaps.

Within settlement categories, urban POIs tend to exhibit slightly higher consistency across supports, whereas forested and upland locations show greater divergence between supports. This reflects differences in spatial heterogeneity, where mixed land cover within aggregation windows introduces additional variability in completeness and stability metrics.

These detailed POI-level results provide additional context for interpreting completeness thresholds by highlighting how similar summary statistics can correspond to different temporal structures of observation gaps.

# VII. Settlement-Scale Observability Summary

***Table S3.*** *Settlement-scale observability metrics including spatial completeness, temporal completeness under varying thresholds, dropout duration, pixel share, and radiance variability (CV) across GHSL SMOD classes.*

| **Settlement Class** | **Pixel Share (%)** | **SC (%)** | **TC (%)** | | | **Maximum Dropout Duration (Days)** | **Temporal CV** | **Spatial CV** |
|---|---|---|---|---|---|---|---|---|
| | | | *≥ 25%* | *≥ 50%* | *≥ 75%* | | | |
| **Urban Centre (30)** | **0.55** | **36.6** | **47.4** | **35.9** | **24.7** | 40 | **1.56** | **1.16** |
| **Urban Cluster** | **14.39** | **38.6** | **50.7** | **37.9** | **24.2** | 45 | **1.23** | **1.42** |
| Dense Urban Cluster (23) | 11.14 | 37.9 | 50.6 | 37.9 | 23.9 | 45 | 2.03 | 1.49 |
| Semi-dense Urban Cluster (22) | 1.57 | 39.1 | 51.1 | 38.4 | 24.6 | 33 | 0.81 | 1.17 |
| Suburban/Peri-urban Cluster (21) | 1.68 | 38.9 | 50.6 | 37.3 | 24.0 | 33 | 0.85 | 1.62 |
| **Non-urban Cells** | **84.15** | **38.2** | **52.5** | **39.0** | **23.5** | **45** | **1.36** | **1.28** |
| Rural Cluster (13) | 63.78 | 39.5 | 53.4 | 39.8 | 25.0 | 33 | 2.18 | 1.43 |



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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Low-density Rural (12) | 16.39 | 37.7 | 51.8 | 38.6 | 22.6 | 45 | 1.07 | 1.33 |
| Very Low-density Rural (11) | 3.98 | 37.6 | 52.4 | 38.7 | 22.8 | 33 | 0.83 | 1.08 |
| **Water (10)** | **0.99** | **66.6** | **57.8** | **47.7** | **38.8** | **26** | **7.49** | **1.41** |

Table S3 reports settlement-scale observability metrics across the Global Human Settlement Layer Degree of Urbanisation Settlement Model classes (GHSL SMOD; Schiavina et al., 2023). This includes spatial completeness (SC), temporal completeness under admissibility thresholds (≥25%, ≥50%, ≥75%), maximum dropout duration, pixel share, and radiance variability (temporal and spatial CV).

While spatial completeness varies only slightly across settlement types, the response to increasing admissibility thresholds differs between classes. Urban and peri-urban categories show relatively uniform declines in temporal completeness as thresholds increase, whereas rural subclasses exhibit more variable responses, reflecting differences in spatial fragmentation and baseline signal strength.

Radiance variability further differentiates subclasses within broader settlement groups. Dense urban clusters exhibit higher temporal CV relative to aggregated urban clusters, indicating greater intra-class heterogeneity despite higher overall brightness. In contrast, very low-density rural classes show lower temporal variability but also reduced signal magnitude, limiting interpretability despite apparent stability.

Pixel share strongly conditions the aggregate representation of these metrics. Non-urban classes dominate the spatial domain, meaning that unstratified analyses would disproportionately reflect lower-intensity, higher-noise environments. This reinforces the role of settlement stratification in isolating infrastructure-relevant signals from background variability.

Taken together, the settlement-scale metrics highlight that observability is not only a function of completeness, but also of how variability, spatial composition, and threshold sensitivity interact across settlement regimes.

## VIII. Example GHSL Cumulative Thematic Masks Used for Aggregation

Figure S6 illustrates representative GHSL SMOD cumulative thematic masks used for aggregation of NTL radiance. These masks define spatial subsets corresponding to distinct settlement regimes, ranging from dense urban cores to rural and non-urban classes.

The masks differ not only in spatial extent but also in morphological structure. Urban masks are spatially compact and concentrated around high-intensity emission centres, whereas rural and non-urban masks are more spatially dispersed and fragmented. This structural contrast influences both the density of radiance signals and the distribution of valid observations within each class.



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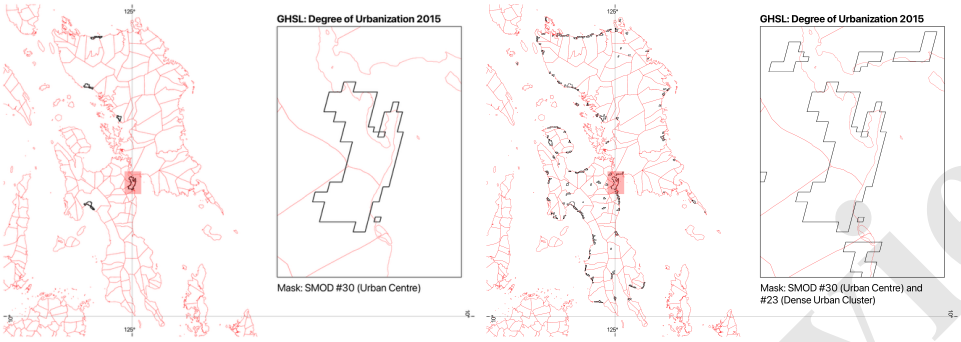


***Figure S6**. Representative cumulative thematic masks derived from GHSL SMOD classes, showing G1 (Urban Core, code 30) and G2 (Dense Urban, codes 23–30).*

Differences in mask composition also affect the degree of spatial mixing during aggregation. Urban masks primarily capture contiguous built-up areas, while broader masks incorporate heterogeneous land covers, including transitional and low-emission zones. These distinctions provide the basis for evaluating how settlement structure conditions both signal stability and alignment with external datasets.

## IX. Effect of Spatial Completeness Threshold on Event Signal Structure

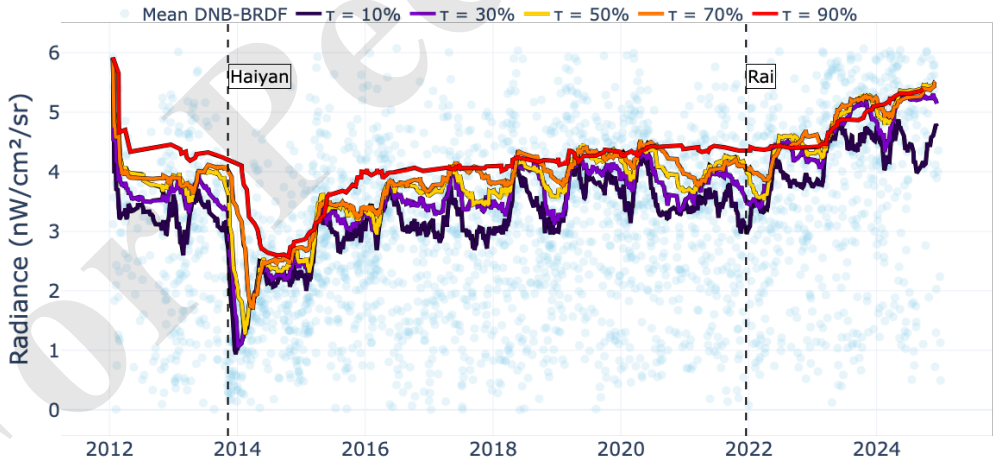


***Figure S7.** NTL time series under varying spatial completeness thresholds (τ), showing changes in short-term variability and temporal continuity during the Typhoon Haiyan period.*

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Figure S7 illustrates the effect of varying spatial completeness thresholds (τ) on the structure of NTL time series during the Typhoon Haiyan period. Time series are shown for multiple admissibility levels, highlighting how threshold selection alters the representation of event-driven radiance dynamics.

Lower thresholds retain a larger number of observations but introduce increased variability and residual noise, resulting in less distinct event signatures. As τ increases, short-term fluctuations are reduced, but the time series becomes progressively fragmented as stricter thresholds exclude a larger proportion of dates. However, this increased clarity is accompanied by fragmentation of the time series, as stricter thresholds exclude a substantial fraction of days.

The effect of thresholding extends beyond noise reduction. Changes in τ alter the apparent depth, duration, and temporal continuity of the radiance decline associated with the event. At lower thresholds, the disruption signal may appear attenuated or temporally diffuse, while at higher thresholds, the signal becomes sharper but discontinuous due to missing segments. This demonstrates that threshold selection directly influences the interpretability of event-scale dynamics.

Thresholding produces a continuous trade-off rather than a universally optimal intermediate setting. Lower thresholds retain more dates but include observations with weaker spatial support, whereas stricter thresholds reduce temporal coverage and increase fragmentation. The preferred threshold therefore depends on the settlement mask, validation target, and relative weighting assigned to association, error, and temporal coverage. The threshold-dependent patterns in Figure S7 occur across settlement classes, although their magnitude and stability vary with baseline radiance and spatial heterogeneity.

## X. Effect of Spatial Completeness Threshold on NTL–Load Signal Structure

This section examines how spatial completeness thresholds (τ) alter settlement-stratified VNP46A2 radiance and its correspondence with NGCP sub-grid electricity load (Román et al., 2018; National Grid Corporation of the Philippines, n.d.). Figure S8 presents unnormalised DNB–BRDF radiance across GHSL classes under representative thresholds (τ = 10%, 50%, and 90%), alongside NGCP load.

At τ = 10%, temporal coverage is highest and the urban-focused series retain strong correspondence with NGCP load, although greater short-term variability remains visible. At τ = 50%, the broad temporal trajectory is preserved but fewer dates contribute to the series. At τ = 90%, retained observations are sparse and the time series becomes highly fragmented, limiting representation of continuous system dynamics. The comparison demonstrates that increasing τ does not uniformly improve alignment; it progressively exchanges temporal coverage for stronger spatial support.



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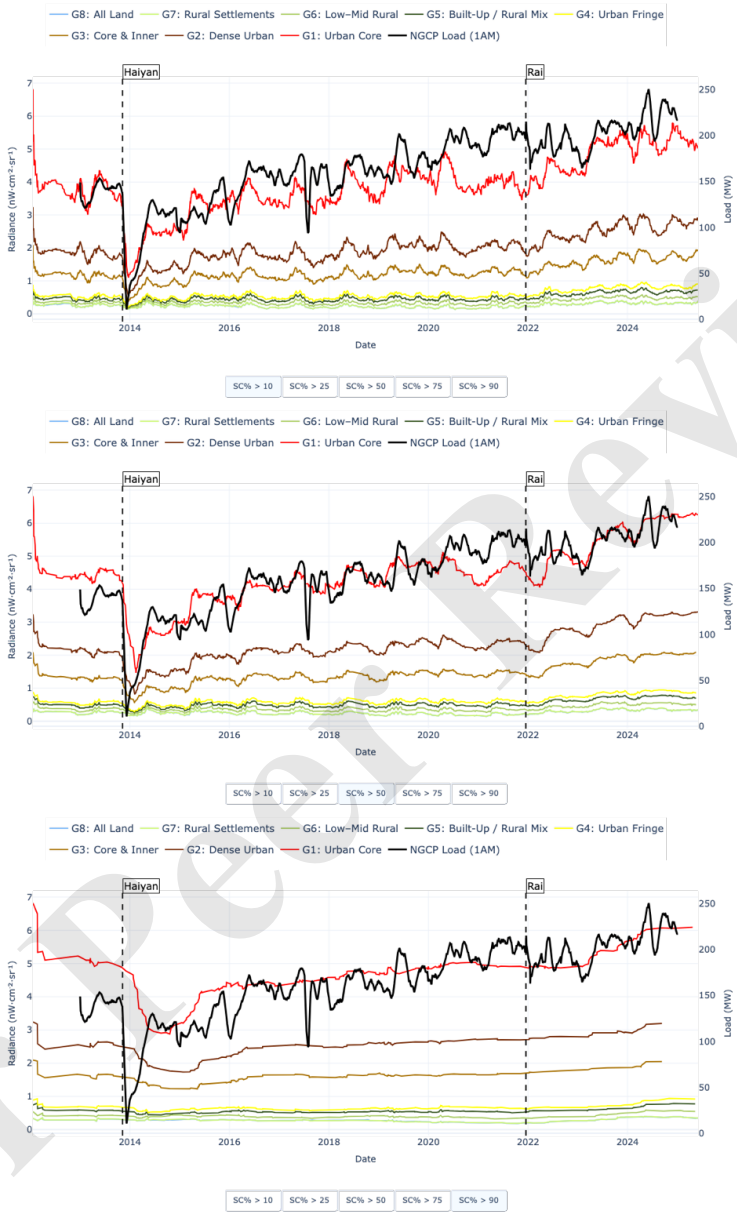


***Figure S8****. GHSL-stratified DNB–BRDF radiance and NGCP load under varying spatial completeness thresholds: (a) τ = 10%, (b) τ = 50%, (c) τ = 90%. Dashed vertical lines indicate major typhoon events (Haiyan and Rai).*

Across all thresholds, the effect is most pronounced in urban classes, where baseline radiance is highest and alignment with load is strongest. Rural-dominant classes remain comparatively stable but exhibit weaker correspondence with load across all τ values. These results demonstrate



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that threshold selection directly reshapes the amplitude, continuity, and timing of NTL signals, rather than simply filtering noise. The main analysis therefore evaluates this trade-off using the composite score rather than prespecifying an intermediate threshold as universally preferable.

## XI. Top Five Reliability-Qualified Alignment Configurations for Each Visayas Sub-Grid

*Table S10. Five highest-scoring reliability-qualified NTL–load alignment configurations within each Visayas sub-grid. Composite scores combine within-sub-grid normalised Pearson correlation, inverted RMSE, and retained temporal coverage using weights of 0.6, 0.1, and 0.3, respectively. Scores rank configurations within each sub-grid and should not be compared directly between sub-grids.*

| Sub-grid | Rank | Thematic Mask | GHSL Range | τ (%) | Composite Score | r | RMSE | Temporal Coverage (%) |
|---|---|---|---|---|---|---|---|---|
| **Samar–Leyte** | **1** | **G2: Dense Urban** | **23-30** | **10** | **0.937** | **0.84** | **0.57** | **49.81** |
| | 2 | G1: Urban Core | 30 | 10 | 0.934 | 0.85 | 0.54 | 46.59 |
| | 3 | G3: Core & Inner | 22-30 | 10 | 0.922 | 0.82 | 0.61 | 50.25 |
| | 4 | G1: Urban Core | 30 | 20 | 0.907 | 0.86 | 0.53 | 40.82 |
| | 5 | G2: Dense Urban | 23-30 | 20 | 0.907 | 0.85 | 0.55 | 42.86 |
| **Bohol** | **1** | **G1: Urban Core** | **30** | **20** | **0.880** | **0.86** | **0.54** | **28.71** |
| | 2 | G1: Urban Core | 30 | 10 | 0.875 | 0.85 | 0.56 | 31.32 |
| | 3 | G1: Urban Core | 30 | 30 | 0.860 | 0.86 | 0.54 | 26.62 |
| | 4 | G2: Dense Urban | 23-30 | 20 | 0.859 | 0.83 | 0.57 | 32.50 |
| | 5 | G3: Core & Inner | 22-30 | 30 | 0.859 | 0.84 | 0.56 | 30.45 |
| **Cebu** | **1** | **G1: Urban Core** | **30** | **60** | **0.735** | **0.78** | **0.67** | **21.38** |
| | 2 | G1: Urban Core | 30 | 50 | 0.728 | 0.77 | 0.68 | 23.57 |
| | 3 | G1: Urban Core | 30 | 30 | 0.717 | 0.75 | 0.71 | 27.53 |
| | 4 | G2: Dense Urban | 23-30 | 70 | 0.712 | 0.78 | 0.66 | 15.82 |
| | 5 | G1: Urban Core | 30 | 40 | 0.708 | 0.75 | 0.70 | 25.51 |
| **Negros** | **1** | **G1: Urban Core** | **30** | **10** | **0.897** | **0.78** | **0.66** | **38.69** |
| | 2 | G2: Dense Urban | 23-30 | 10 | 0.881 | 0.72 | 0.75 | 42.97 |
| | 3 | G1: Urban Core | 30 | 20 | 0.867 | 0.79 | 0.65 | 33.64 |
| | 4 | G3: Core & Inner | 22-30 | 10 | 0.862 | 0.68 | 0.81 | 43.83 |
| | 5 | G2: Dense Urban | 23-30 | 20 | 0.857 | 0.75 | 0.71 | 36.14 |
| **Panay** | **1** | **G2: Dense Urban** | **23-30** | **10** | **0.889** | **0.80** | **0.64** | **43.89** |
| | 2 | G2: Dense Urban | 23-30 | 20 | 0.877 | 0.84 | 0.56 | 37.30 |
| | 3 | G3: Core & Inner | 22-30 | 10 | 0.873 | 0.77 | 0.68 | 44.48 |
| | 4 | G3: Core & Inner | 22-30 | 20 | 0.865 | 0.82 | 0.61 | 38.33 |
| | 5 | G2: Dense Urban | 23-30 | 30 | 0.853 | 0.85 | 0.54 | 32.50 |

The expanded rankings show that all top-five configurations remain restricted to the three most urban-focused cumulative thematic masks (G1–G3). Within each sub-grid, several neighbouring mask–threshold configurations yield closely comparable scores, although the preferred threshold range differs regionally. Samar–Leyte, Negros, and Panay favour low thresholds, while Bohol favours low-to-moderate thresholds and Cebu favours a broader set of moderate-to-strict G1 configurations. Settlement-mask selection is therefore more stable across sub-grids than the precise completeness threshold.



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# Highlights

**Reliability Conditions for Interpreting Daily NASA Black Marble Nighttime Lights in Cloud-Impacted Tropical Regions**

Rene Jr. L. Principe,Karin Reinke,Simon Jones

- Daily Black Marble records in the tropics are frequently interrupted by multi-week cloud-driven gaps.
- A diagnostic-first framework quantifies completeness, dropout duration, and radiance stability before downstream interpretation.
- At a 50% spatial completeness threshold, only 36–39% of days remain admissible across land-based settlement classes.
- Urban-focused GHSL masks produce the strongest NTL–load alignment across five sub-grids ($r = 0.78$–$0.86$).
- The highest-scoring spatial completeness threshold varies among sub-grids, supporting transferable selection logic rather than a universal threshold.

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# Reliability Conditions for Interpreting Daily NASA Black Marble Nighttime Lights in Cloud-Impacted Tropical Regions

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ABSTRACT

Satellite-derived nighttime lights (NTL) are widely used to infer electricity disruption and recovery following disasters, yet the conditions under which daily radiance can be interpreted reliably remain fundamentally unresolved, particularly in persistently cloud-affected tropical regions. Although the VIIRS Day/Night Band and NASA's Black Marble (VNP46A2) product provide radiometrically calibrated near-daily observations, cloud obstruction, quality-flag filtering, and temporal interpolation introduce irregular sampling and uneven data completeness, limiting their reliability for time-series analyses of electricity disruption and recovery.

This study develops a diagnostic-first framework to quantify the spatiotemporal observability of daily Black Marble radiance prior to outage analysis. Eastern Visayas in the Philippines, which experienced severe grid collapse during Typhoon Haiyan (2013), is used as an extreme stress-test environment. Spatial completeness, temporal completeness, maximum dropout duration, and settlement-stratified radiance stability are evaluated at pixel and settlement scales. Reliability-qualified radiance trajectories are then aligned with independent sub-grid electricity load to identify admissible completeness thresholds under which daily NTL most faithfully track infrastructure dynamics. Results show that daily NTL observations in humid tropical environments are intermittently available, with frequent extended cloud-driven observation gaps rather than continuous daily records. Spatial aggregation improves signal stability but does not recover cloud-obscured observations. Alignment consistently favours urban-focused masks, while the highest-scoring spatial completeness threshold varies among sub-grids. Transferability across neighbouring Visayas sub-grids indicates that these conditions reflect underlying observational constraints rather than exclusively site-specific effects.

By embedding explicit reliability diagnostics upstream of impact inference, this study establishes reproducible conditions under which daily NTL can support electricity monitoring in cloud-prone, hazard-exposed regions. The framework provides a methodological foundation for more robust outage detection and recovery assessment using satellite-derived nighttime lights.

## 1. Introduction

Satellite-derived nighttime lights (NTL) provide one of the few globally consistent, spatially continuous observational records of electrified human activity. Artificial illumination observed from space has been repeatedly shown to correlate with settlement presence, population distribution, infrastructure intensity, and economic output (Elvidge et al., 1997; Chen and Nordhaus, 2011; Henderson et al., 2012). Since artificial lighting depends fundamentally on electricity access and consumption, satellite-measured nocturnal radiance offers a scalable proxy for the functioning of energy-dependent human systems. Unlike many socio-economic datasets that are spatially incomplete, administratively inconsistent, or temporally delayed, NTL observations are acquired systematically at global scale and have proven particularly valuable in data-constrained contexts where official statistics are sparse or lagged (Jean et al., 2016; Tingzon et al., 2019; Zheng et al., 2023; Levin, 2025).

The systematic use of NTL in Earth observation began with the Defense Meteorological Satellite Program's Operational Linescan System (DMSP–OLS). Early studies demonstrated that DMSP nightlight composites could delineate urban extents and approximate the spatial distribution of human settlements (Elvidge et al., 1997; Small et al., 2005). Subsequent analyses leveraged DMSP data to estimate economic production and growth patterns (Chen and Nordhaus, 2011; Henderson et al., 2012) and to track multi-decadal urban expansion (Small et al., 2005). DMSP-derived radiance was also applied in disaster contexts, including tsunami impact assessment in Southeast Asia (Gillespie et al., 2014) and analyses of settlement proximity shifts following major flood events (Mård et al., 2018). However, DMSP–OLS suffered from well-documented limitations, including coarse spatial resolution (~2.7 km), 6-bit radiometric quantization, saturation in bright urban cores, blooming artifacts, and reliance on annual composites that suppressed short-term variability (Small et al., 2005). These constraints limited its suitability for detecting rapid infrastructure disruption or localized power loss. The launch of the Visible Infrared Imaging Radiometer Suite (VIIRS) aboard the Suomi National Polar-orbiting Partnership satellite marked a generational transition in nighttime remote sensing. The VIIRS Day/Night Band (DNB) provides radiometrically calibrated measurements with substantially improved dynamic range, enhanced stray-light correction, and consistent near-global daily coverage (Miller et al., 2013; Kyba et al., 2015). This

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advancement transformed NTL from an annually aggregated indicator into a temporally resolved measurement capable of capturing day-to-day changes in electrified systems. Daily VIIRS observations have since been used to monitor disaster-induced outages (Zhao et al., 2018, 2020), track grid restoration (Román et al., 2019), and quantify societal disruption during the COVID-19 pandemic (Elvidge et al., 2020; Ghosh et al., 2020; Stokes and Román, 2022).

NASA's Black Marble Product Suite (VNP46 series) further refined NTL observations by introducing physically based corrections for bidirectional reflectance distribution function (BRDF) effects, lunar irradiance, atmospheric scattering, and sensor geometry (Román et al., 2018). In addition to these radiometric adjustments, Black Marble provides per-pixel quality flags and a gap-filling framework intended to mitigate cloud-induced data loss. These developments enabled increasingly quantitative applications of daily NTL for infrastructure disruption and recovery analysis. However, the observational reliability of daily NTL remains unevenly treated across the literature. Here, *observability* refers to the extent to which radiance values correspond to direct clear-sky surface measurements rather than temporally propagated or interpolated estimates. The information content of a daily time series therefore remains dependent on atmospheric conditions, quality screening, spatial support, and preprocessing design. No widely adopted framework currently integrates these factors into explicit admissibility conditions before disaster-related interpretation (Zhao et al., 2019; Zheng et al., 2023; Levin, 2025).

The availability of daily observations does not, however, imply that the resulting radiance series records only changes in human activity. VIIRS DNB radiance remains sensitive to viewing geometry, natural background illumination, atmospheric conditions, and spatially varying surface structure. Multi-temporal analyses have shown systematic anisotropy in artificial-light observations as satellite view angle changes, particularly within vertically complex urban environments (Li et al., 2019). Natural-light variation can also affect the detection of faint artificial sources, although background-based corrections substantially reduce this variability (Coesfeld et al., 2020). More broadly, retrieval-level uncertainty remains associated with atmospheric correction, viewing geometry, surface reflectance, and cross-sensor consistency even after operational processing (Wang et al., 2021). Tan et al. (2023) further demonstrated that environmental and observational factors explain spatially varying proportions of daily radiance variability and that these effects differ between urban and rural settings. These studies show that radiometric stability and observation availability are related but distinct dimensions of NTL reliability.

Existing studies address irregular daily observations through several distinct strategies. One is temporal aggregation, which exchanges event-scale detail for a more stable representation of longer-term conditions. NASA's VNP46A3 and VNP46A4 products combine quality-screened VNP46A2 observations into monthly and annual composites stratified by viewing-angle category and snow condition, while retaining quality and observation-count information (Wang et al., 2022b). Such products improve temporal representation for long-term and structural analyses, but necessarily change the temporal scale at which variation can be resolved. Zheng et al. (2022a) showed that alternative temporal compositing strategies can materially alter NTL magnitudes, spatial distributions, temporal trends, and relationships with downstream variables. The use of a temporally aggregated or smoothed series therefore does not simply remove noise; it defines which forms and durations of change remain observable. This distinction is consequential for disaster applications, where electricity disruption and early recovery may occur within the period absorbed by a monthly composite.

Empirical studies illustrate considerable methodological variation in preprocessing workflows, treatment of observation gaps, and approaches to assessing data reliability. Román et al. (2019) demonstrated a strong correspondence between BRDF-corrected radiance and electricity load after Hurricane Maria, establishing the credibility of Black Marble for the tracking of restoration at the territory level. Subsequent work has expanded methodological sophistication in modeling and scaling, including machine-learning approaches to gap-filled time series (Chakraborty and Stokes, 2023), gap-aware decomposition techniques (Jia et al., 2023), and sensitivity analyses of recovery-duration estimates (Zheng et al., 2025).

Electricity-focused applications have further strengthened the empirical basis for interpreting NTL as an indicator of power-system conditions across different spatial and temporal scales. Chakraborty et al. (2025) used a decade of Black Marble observations to map global electricity access and persistent electrification gaps at 1 km resolution, while Yang et al. (2026) integrated NTL, prefecture-level electricity consumption, and GHSL classes to produce settlement-stratified consumption estimates at 500 m resolution. At shorter temporal scales, Li and Wang (2026) related monthly VIIRS-derived dispersion, anomaly, and seasonal indicators to electricity availability, outage shares, and load-shedding conditions in South Africa. Xu et al. (2026) developed a self-adaptive anomaly-detection framework for daily NTL and derived a satellite-based index that was evaluated against ground-measured grid reliability. For disaster recovery, Mo et al. (2025) estimated blackout durations following 396 tropical cyclones across 66 countries, validated NTL-derived durations against PowerOutage.US records, and represented periods without valid observations as uncertainty intervals. Collectively, these studies establish that NTL can represent electricity access, consumption, reliability, outages, and recovery, but they do so using different temporal products, validation targets, and processing assumptions (Zhao et al., 2019; Levin, 2025). Proxy relevance and observational admissibility must therefore be treated as separate questions.

A parallel body of work treats unavailable observations primarily as a reconstruction problem. Hao et al. (2023)



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combined spatiotemporal weighting with bidirectional recurrent modelling to predict missing daily Black Marble values, while Tan and Zhu (2023) developed the CRYSTAL method to reconstruct cloud-obscured daily NTL by integrating spatial and temporal information from partially clear imagery. In the wider optical remote-sensing literature, spatial-spectral random forests similarly exploit neighbouring and multispectral information to predict cloud-obscured pixels (Wang et al., 2022a). These approaches make important contributions by improving the completeness and predictive accuracy of reconstructed image sequences. Their objective nevertheless differs from determining whether the original directly observed record provides sufficient support for a specific inference. Reconstruction produces model-derived values for unavailable dates, whereas observability concerns the direct measurements supporting interpretation. A reconstructed sequence may therefore be temporally complete without being continuously observed.

These methodological distinctions are most consequential in persistently cloudy regions, where valid DNB-BRDF retrievals are irregular in both space and time. In humid tropical and monsoonal climates, coherent cloud systems can reduce valid-pixel support across extensive areas and produce prolonged observation gaps (Zhao et al., 2019; Skoufias et al., 2021). As a result, the number of directly observed measurements varies substantially across space and time. Recent disaster applications reflect different strategies for coping with this constraint. Some studies restrict analysis to a limited number of clear-sky observations to preserve interpretability (Hu et al., 2025). Others rely on multi-day or monthly composites to extract seasonal or structural patterns, but at the cost of resolving short-lived outages or rapid recovery dynamics (Mokhtari et al., 2025). Although Black Marble's gap-filling procedure improves nominal continuity (Román et al., 2018), it does not increase the number of directly observed clear-sky measurements. Consequently, hazard-impact inference remains sensitive to cloud persistence, filtering thresholds, and aggregation design (Levin, 2025; Zheng et al., 2023).

Recent applications illustrate how observation limitations are commonly handled within event-specific preprocessing rather than evaluated as a primary result. For the 2023 Türkiye earthquakes, Levin (2023) retained cloud-free VNP46A1 observations, compared pre- and post-event periods of different lengths, and reported dates for which no cloud-free pixels were available. The study demonstrated that carefully selected nightly observations can map acute earthquake impacts and recovery. Its event-specific design did not seek to establish transferable criteria for the amount or continuity of direct observation required for interpretation. Similarly, Zheng et al. (2022b) used daily VNP46A2 radiance to identify conflict-related light reductions and estimate populations exposed to power shortages, while recognising background noise and the absence of field validation as limitations. COVID-19 studies used temporally aggregated VIIRS observations to detect sustained reductions in urban activity (Elvidge et al., 2020), demonstrating the value of temporal smoothing for persistent changes but not necessarily for interruptions lasting only several days. These applications demonstrate the substantive value of NTL, but observational support is generally managed through date selection, temporal aggregation, uncertainty representation, or application-specific screening. It is less commonly evaluated as a separate upstream condition using a shared set of diagnostics.

In addition to atmospheric constraints, settlement heterogeneity influences both the stability and interpretation of radiance. Dense urban centres typically exhibit persistent, high-intensity illumination, whereas peri-urban and rural settlements often display weaker signals and greater relative fluctuation (Small et al., 2005). The relationship between radiance and the human processes it represents is also spatially non-stationary. Wu et al. (2022) found that roads, residential areas, commercial activity, and other urban features contributed differently to VIIRS radiance across locations and spatial scales. Martinez et al. (2023) likewise showed that the suitability of daily gap-filled Black Marble radiance for population analysis varied with spatial and temporal aggregation. Together with the spatially heterogeneous uncertainty identified by Tan et al. (2023), these findings show that apparent radiance changes cannot be interpreted independently of settlement structure and analytical scale.

Taken together, the literature reveals a methodological tension. Daily NTL are increasingly used as near-real-time indicators of electricity disruption and recovery, yet the underlying observations are variously filtered, aggregated, smoothed, reconstructed, or modelled before interpretation. These procedures can improve radiometric stability, temporal representation, or predictive continuity, but they do not answer the same question. Few studies jointly assess whether the direct observation record provides sufficient spatial support, temporal representation, and signal stability for the intended analysis. This limitation is particularly consequential in tropical cyclone regions, where severe infrastructure disruption coincides with persistent cloud cover.

Existing studies therefore recognise data-quality limitations, but generally address them within application-specific preprocessing. What remains less developed is a common upstream framework for translating these dimensions into explicit admissibility conditions. In particular, spatial and temporal completeness, maximum dropout duration, and settlement-dependent variability are rarely examined together as explicit admissibility diagnostics.

A preliminary version of this work introduced the reliability-qualified NTL concept and demonstrated initial NTL–electricity load alignment in the Visayas (Principe et al., 2026). The present study substantially extends that contribution by formalising the observability metrics, adding pixel- and settlement-scale diagnostics, evaluating long-term and global completeness patterns, expanding the cross-sub-grid assessment, and providing a systematic framework for selecting admissibility conditions.

To address this gap, this study develops a diagnostic-first framework for evaluating daily Black Marble observations in



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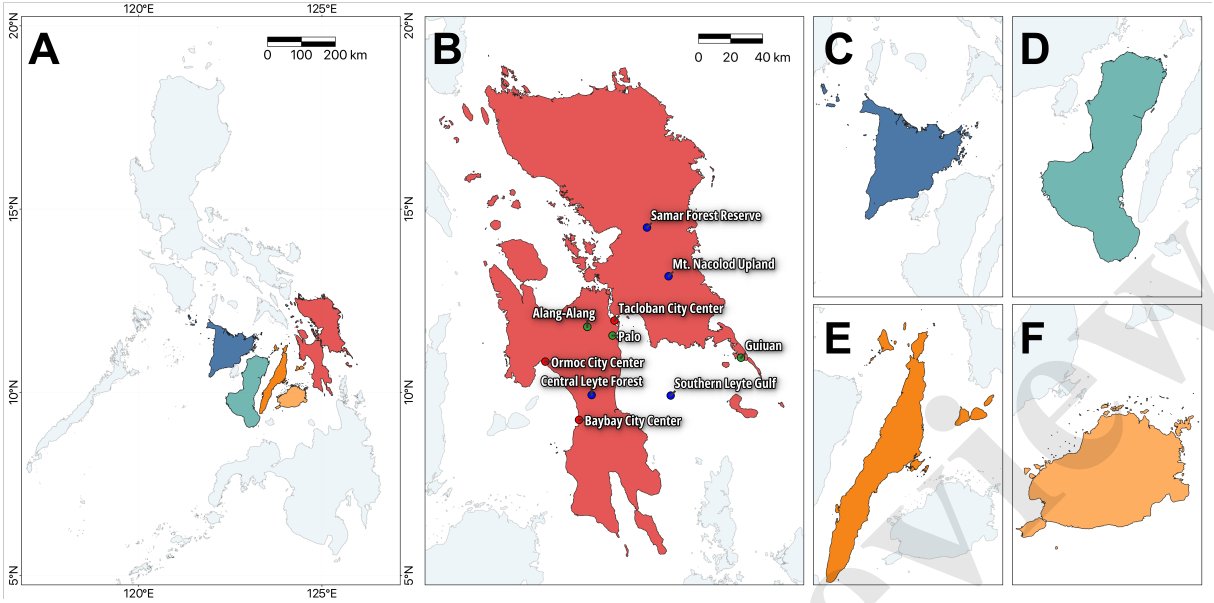


**Figure 1:** Study region and analysis units. (A) Locations of the five National Grid Corporation of the Philippines (NGCP) Visayas sub-grid service areas. (B) Samar–Leyte, the primary diagnostic region, with the ten points of interest (POIs) used for pixel-scale observability analysis. (C–F) Panay, Negros, Cebu, and Bohol sub-grids, respectively, used for cross-sub-grid NTL–load alignment assessment.

a cloud-prone tropical environment. The framework quantifies observation availability, gap duration, and radiance stability across pixel and settlement scales before aligning reliability-qualified DNB-BRDF trajectories with independent sub-grid electricity load. It then applies the same assessment across five Visayas sub-grids to determine which reliability patterns are consistent and which remain context dependent. The resulting framework establishes empirical conditions under which daily NTL can be interpreted more defensibly and provides a foundation for subsequent outage, recovery, and resilience analysis.

## 2. Materials and Methods

### 2.1. Study Region

The empirical focus of this study is the Samar–Leyte island group in Eastern Visayas, Philippines (Figure 1). The region represents an extreme test environment for daily NTL analysis due to the concurrence of persistent tropical cloud systems and severe electricity disruption during Super Typhoon Haiyan (November 2013). Haiyan was one of the most powerful tropical cyclones ever recorded at landfall, affecting over 16 million people and causing widespread infrastructure collapse across Eastern Visayas (National Disaster Risk Reduction and Management Council, 2014; United Nations Office for Disaster Risk Reduction, 2015). The event resulted in prolonged electricity outages, transmission failures, and system fragmentation across the Samar–Leyte sub-grid.

More broadly, the Philippines is consistently ranked among the most disaster-exposed countries globally due to its geographic location along the Western Pacific typhoon belt and the Pacific Ring of Fire (Bündnis Entwicklung Hilft and Institute for International Law of Peace and Armed Conflict (IFHV), 2023; United Nations Office for Disaster Risk Reduction, 2022). Frequent cyclones, high rainfall variability, and complex islanded grid structures combine to create persistent challenges for optical satellite observability and infrastructure monitoring. These conditions make Samar–Leyte a stringent stress test for evaluating when daily NTL can reliably reflect electricity-related dynamics.

Figure 1 situates the five Visayas sub-grids within the Philippine archipelago (Fig. 1A). Samar–Leyte, shown in Fig. 1B, contains ten points of interest (POIs) classified as city centres (red), municipal centres (green), and reference sites (blue). The city-centre POIs are Tacloban, Ormoc, and Baybay; the municipal-centre POIs are Palo, Alang-Alang, and Guiuan; and the reference sites are Samar Forest Reserve, Mt. Nacolod Upland, Central Leyte Forest, and Southern Leyte Gulf. Detailed site descriptions, coordinates, and selection rationale are provided in Table S1. These POIs were used to examine pixel-scale NTL behaviour across contrasting illumination contexts and spatial supports before settlement-scale aggregation.

Regional consistency is evaluated across Panay, Negros, Cebu, and Bohol, shown in Fig. 1C–F, respectively. These sub-grids were selected because independent electricity load data are available at the corresponding operational scale. The same reliability-qualified alignment framework is applied to each sub-grid to assess how the correspondence between daily NTL and electricity load varies across island



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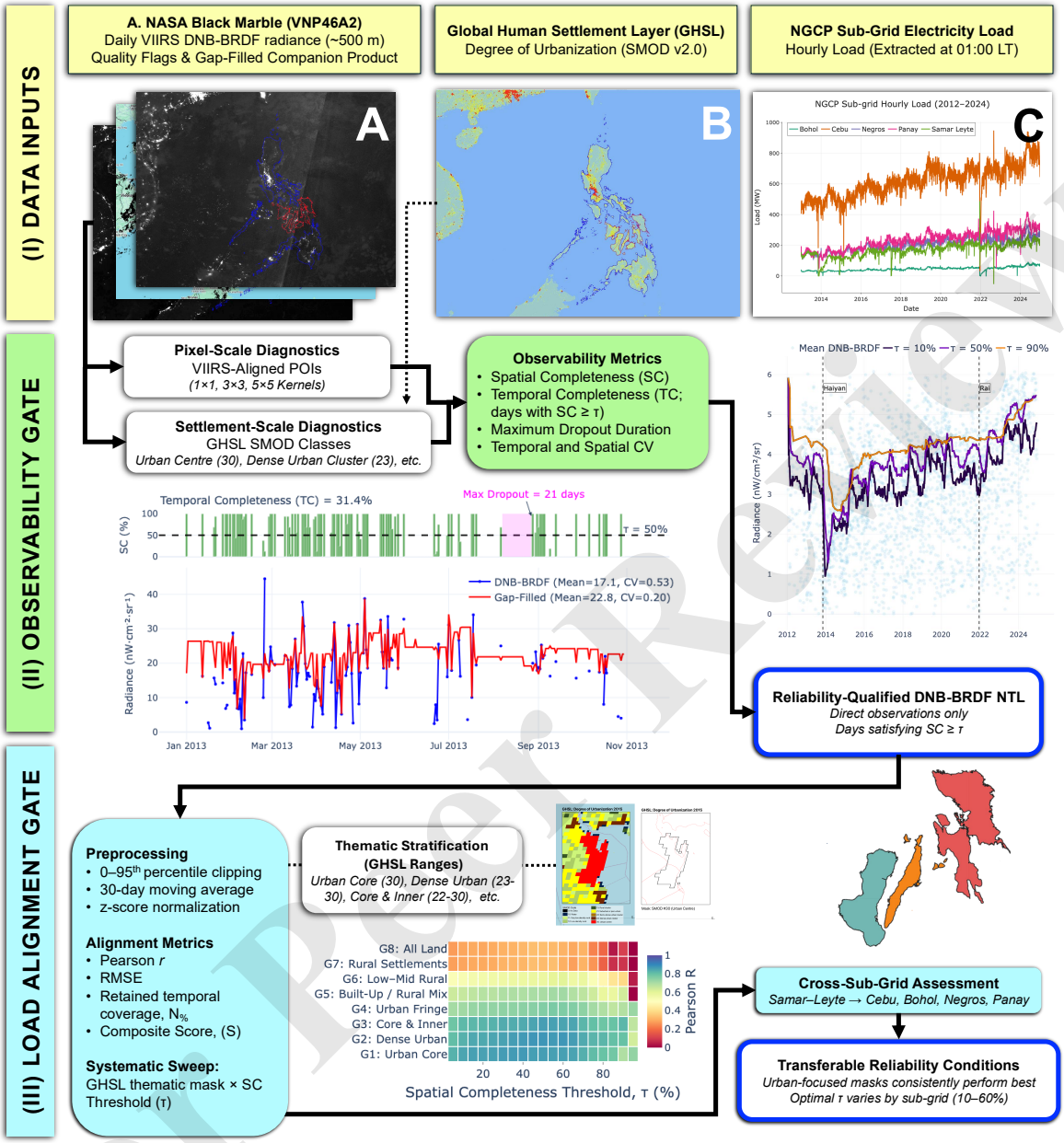


**Figure 2:** Overview of the reliability-qualified NTL framework. Inputs comprise (A) NASA Black Marble VNP46A2 radiance, (B) Global Human Settlement Layer settlement classes, and (C) NGCP sub-grid electricity load. The workflow progresses from (i) data inputs to (ii) pixel- and settlement-scale observability screening and (iii) reliability-qualified NTL–load alignment and cross-sub-grid assessment.

morphologies, settlement structures, and observational conditions.

## 2.2. Datasets

The analytical design integrates three data streams corresponding to Fig. 2: (A) NASA Black Marble VNP46A2 radiance, (B) Global Human Settlement Layer (GHSL) settlement morphology, and (C) National Grid Corporation of the Philippines (NGCP) sub-grid electricity load. Datasets A and B support the observability diagnostics, while all three are integrated for reliability-qualified NTL–load alignment.

Daily NTL observations (Fig. 2A) are derived from NASA's Black Marble VNP46A2 product (Collection 2, Version 1) at approximately 500 m spatial resolution. Two radiance layers are analysed: the BRDF- and lunar-corrected



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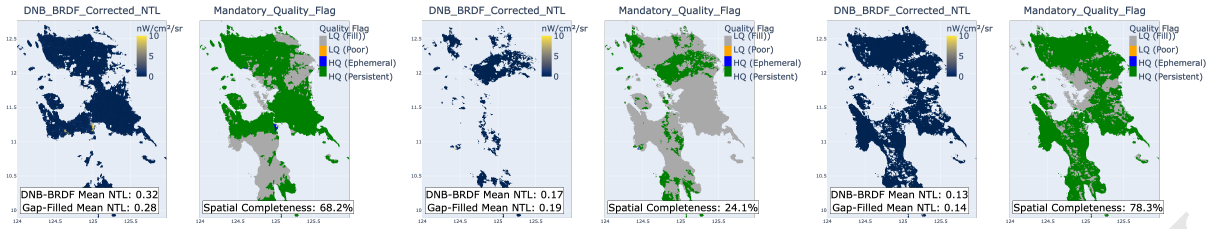


(a) 1 Nov 2013 (b) 5 Nov 2013 (c) 10 Nov 2013

**Figure 3:** Daily DNB–BRDF radiance and corresponding quality flags for 1, 5, and 10 November 2013, bracketing the 8 November landfall of Typhoon Haiyan. Annotated values indicate mean radiance and spatial completeness (SC), highlighting day-to-day variability in valid-pixel coverage under persistent cloud conditions.

Day/Night Band radiance (DNB-BRDF) and its spatiotemporally interpolated counterpart (Gap-Filled). VNP46A2 tiles intersecting the Samar–Leyte, Bohol, Cebu, Negros, and Panay sub-grid service areas were accessed through Google Earth Engine (asset: `NASA/VIIRS/002/VNP46A2`).

Direct DNB-BRDF observations were retained only where the `Mandatory_Quality_Flag` equalled 0 and the corresponding `QF_Cloud_Mask` conditions satisfied the adopted clear-sky screening rule. All remaining pixels, including low-quality, cloud-contaminated, and fill observations, were assigned as unavailable. The Gap-Filled band is retained strictly for comparative diagnostics and is not used to substitute missing observations in reliability-qualified analyses. As illustrated in Fig. 3, daily DNB-BRDF coverage exhibits marked spatial discontinuities under persistent cloud systems, including over dense urban areas. These structural gaps motivate the observability diagnostics developed.

Settlement morphology (Fig. 2B) is characterised using the Global Human Settlement Layer (GHSL) Degree of Urbanisation Settlement Model (SMOD), version P2023A (V2.0), produced by the European Commission Joint Research Centre (Schiavina et al., 2023). GHSL SMOD is provided at 1 km spatial resolution and classifies each grid cell into a globally harmonised hierarchy of settlement types derived from built-up density and population distribution. Because SMOD represents structural urban form rather than short-term change, the 2015 epoch is used as a stable stratification layer for the 2012–2024 NTL analysis window. For diagnostic consistency, SMOD Level-2 classes are grouped into three analytical tiers: Urban Centre (code 30), Urban Cluster (codes 21–23), and Non-urban Cells (codes 11–13), with water (code 10) retained separately (Table 1). These classes complement the pixel-based POIs, enabling cross-scale evaluation of radiance completeness and stability. For NTL–load alignment, the same classes are combined into cumulative GHSL thematic masks ranging from the Urban Core mask (G1: code 30) to the All Land mask (G8: codes 10–30).

Hourly electricity load data (Fig. 2C) were obtained from the NGCP (National Grid Corporation of the Philippines, n.d.) for the Samar–Leyte, Bohol, Cebu, Negros, and Panay operational sub-grids. To align with the VIIRS nighttime overpass at approximately 01:30 local time, daily load values are extracted at 01:00 local time. These sub-grid series provide an independent reference for evaluating the correspondence between reliability-qualified NTL and electricity dynamics.

**Table 1**
GHSL SMOD v2.0 (P2023A) Level-2 classes and analysis groupings used for settlement-stratified diagnostics.

| Code | Official Class | Analysis Group |
|---|---|---|
| -200 | No Data | Excluded |
| 10 | Water | Water |
| 11 | Very low density rural | Non-urban Cells |
| 12 | Low density rural | |
| 13 | Rural cluster | |
| 21 | Suburban or peri-urban | Urban Cluster |
| 22 | Semi-dense urban cluster | |
| 23 | Dense urban cluster | |
| 30 | Urban centre | Urban Centre |

## 2.3. Diagnostic Framework

The methodological workflow is summarised in Figure 2. The framework separates observability diagnostics from electricity alignment to ensure that data-admissibility conditions are established prior to impact inference.

### *2.3.1. Observability diagnostics*

Observability is evaluated for a single reference year, 2013, which represents an extreme stress-test period due to the landfall of Super Typhoon Haiyan and persistent tropical cloud systems. Focusing on this year allows assessment of daily NTL behaviour under simultaneous meteorological obstruction and infrastructure disruption, thereby establishing conservative admissibility conditions.

Two complementary spatial supports are examined to characterise how NTL reliability scales with spatial aggregation. First, pixel-scale diagnostics are conducted at ten pixel-aligned POIs spanning dense urban, peri-urban, forested, and offshore reference environments (Fig. 1; coordinates provided in Supplementary Table S1). Each POI is evaluated

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using nested square kernels (1×1, 3×3, and 5×5 pixels) centred on the VIIRS grid. The progression from single pixels to larger kernels quantifies a scale-dependent trade-off between spatial precision and radiometric stability: small kernels preserve local specificity but are highly sensitive to cloud masking and data gaps, whereas larger kernels increase spatial support and effective sample size, reducing stochastic variability via spatial averaging, albeit at the cost of mixing land uses under heterogeneous cloud cover.

Second, settlement-scale diagnostics are computed using GHSL SMOD v2.0 masks (Table 1). GHSL provides a globally consistent, rule-based Degree of Urbanization classification, avoiding region-specific definitions of "urban" and "rural." Aggregating NTL by settlement class enables evaluation of reliability under structurally distinct illumination regimes while maintaining transferability across regions.

Four observability metrics are computed at each spatial scale: spatial completeness, temporal completeness, maximum dropout duration, and radiance stability.

For an analysis unit $u$ (POI kernel or settlement mask) on day $t$, spatial completeness is defined as

$$SC_{u,t} = \frac{n_{u,t}^{\text{valid}}}{n_u}, \tag{1}$$

where $n_{u,t}^{\text{valid}}$ is the number of valid DNB-BRDF pixels after quality screening and $n_u$ is the total number of pixels in $u$. Spatial completeness therefore measures the effective clear-sky support available for estimating a daily mean radiance and directly reflects cloud obstruction at the unit level. A day is considered admissible if $SC_{u,t} \geq \tau$, where $\tau$ is a valid-pixel threshold. For observability diagnostics, $\tau = 0.5$ (50%) is adopted as a majority-support criterion. This threshold tests whether at least half of the spatial unit is directly observed under clear-sky conditions. Sensitivity analyses across $\tau \in [0.1, 0.9]$ are used to quantify the trade-off between radiometric fidelity and retained temporal coverage.

For analysis unit $u$, year $y$, and spatial completeness threshold $\tau$, temporal completeness is defined as

$$TC_{u,y}(\tau) = \frac{\sum_{t \in y} \mathbb{I}\left(SC_{u,t} \geq \tau\right)}{n_y}, \tag{2}$$

where $\mathbb{I}(\cdot)$ is an indicator function and $n_y$ is the number of calendar days in year $y$. Temporal completeness therefore represents the proportion of days for which the analysis unit satisfies the specified spatial-support requirement. For the 1 × 1 support, this reduces to the proportion of days containing a valid direct observation. Seasonal variability is expected in monsoonal climates; however, the objective is not to model seasonality but to compare completeness distributions across spatial supports under identical atmospheric regimes.

For a specified $\tau$, maximum dropout duration is defined as the longest consecutive sequence of calendar days for which $SC_{u,t} < \tau$. This metric complements temporal completeness by distinguishing dispersed unavailable observations from prolonged uninterrupted gaps that may overlap with disaster disruption or early recovery periods.

Radiance stability is quantified using the coefficient of variation (CV), evaluated after applying the spatial completeness threshold $\tau$. Let $\overline{L}_{u,t}$ denote the daily mean DNB-BRDF radiance for unit $u$ on retained days $\mathcal{T}_u$. Temporal variability is defined as

$$CV_u^{\text{temp}} = \frac{\sigma_{t \in \mathcal{T}_u}(\overline{L}_{u,t})}{\mu_{t \in \mathcal{T}_u}(\overline{L}_{u,t})}, \tag{3}$$

and within-unit spatial variability on retained days is defined as

$$CV_u^{\text{spat}} = \frac{1}{|\mathcal{T}_u|} \sum_{t \in \mathcal{T}_u} \frac{\sigma_{i \in u}(L_{i,t})}{\mu_{i \in u}(L_{i,t})}. \tag{4}$$

These metrics distinguish cross-day volatility from within-day spatial fragmentation. By evaluating CV across kernel sizes and settlement classes under identical $\tau$ thresholds, the framework systematically characterises how spatial aggregation modifies radiance stability under extreme cloud conditions.

Collectively, spatial completeness, temporal completeness, maximum dropout duration, and radiance stability provide a compact representation of daily NTL behaviour. Rather than inspecting individual time series, these metrics summarise the structural limits of observability at the scales where NTL is commonly interpreted.

### *2.3.2. Electricity Load Alignment*

Electricity load alignment evaluates whether reliability-qualified NTL follows independent electricity dynamics across island and island-group sub-grids (e.g., Samar–Leyte, Cebu, Panay; see Figure 2) under varying completeness and aggregation conditions. The objective is not event detection per se, but to determine under which spatial supports and admissibility thresholds daily NTL most consistently tracks independently measured electricity load trends within NGCP sub-grid service areas.

Reliability-qualified NTL series are constructed by aggregating daily DNB-BRDF radiance over unit $u$ using only days satisfying the spatial completeness condition $SC_{u,t} \geq \tau$. Because radiance distributions are positively skewed and occasionally affected by transient artefacts, values above the 95th percentile are winsorised to the 95th-percentile value before smoothing. A centred 30-day moving average is then applied to both NTL and NGCP load series to suppress residual day-level noise and to align structural trends. The 30-day window is intentionally conservative. In tropical settings, cloud-induced gaps may exceed 20 consecutive days, and shorter filters would amplify sampling irregularities. Prior NTL disaster workflows similarly rely on temporal aggregation to reduce cloud-mask omission errors (Román



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et al., 2019), while recent daily NTL modelling studies use smoothed or gap-filled trajectories to stabilise noisy observations (Chakraborty and Stokes, 2023). Finally, both NTL and load are standardised using z-score normalisation. This step removes scale differences between settlement classes (which differ by orders of magnitude in radiance) and between radiance and megawatt load units, allowing alignment to be evaluated in terms of co-movement rather than absolute magnitude.

Alignment is assessed on overlapping admissible dates using three complementary metrics: Pearson correlation coefficient $r$, root mean square error (RMSE), and retained temporal coverage $N_{\%}$. Pearson $r$ measures linear co-variation between settlement-stratified NTL and sub-grid electricity load, while RMSE quantifies their deviation after standardisation. Retained temporal coverage is calculated as the percentage of dates remaining in the common NTL–load analysis period after quality screening, spatial completeness thresholding, and temporal matching. Together, the metrics represent association strength, error magnitude, and temporal representation.

To examine how spatial aggregation influences alignment, a set of cumulative GHSL thematic masks is constructed. In addition to individual SMOD classes (e.g., code 30), cumulative masks are generated (e.g., 23–30, 22–30), progressively expanding from the most intensely illuminated urban cores to include peri-urban and lower-density classes. This design tests a structural hypothesis: electricity load signals are expected to be strongest in dense urban centres where lighting is persistent and electrification is concentrated, while inclusion of dimly lit or sparsely populated classes may dilute signal-to-noise ratios and weaken correspondence with grid-level demand. Evaluating cumulative masks therefore reveals where alignment is maximised and where aggregation begins to introduce instability.

The spatial completeness threshold $\tau$ is varied systematically to quantify the trade-off between radiometric strictness and temporal continuity. Lower $\tau$ values admit more days but risk incorporating spatially fragmented observations; higher $\tau$ values enforce stronger spatial support but reduce retained temporal coverage. For each thematic mask and $\tau$ combination, alignment metrics are computed and visualised as performance surfaces (heatmaps), enabling comparison of performance regions in which association, error, and retained temporal coverage vary jointly. For visual interpretation, the heatmap colour scales are saturated at reference values of $r = 0.80$, RMSE $= 0.60$, and $N_{\%} = 50\%$, with cells meeting the corresponding criterion shown in bold.

Because the mask–threshold sweep produces multiple competing configurations, a composite score is used to summarise the trade-off between alignment, error, and retained temporal coverage:

$$S(u,\tau) = 0.6\,\widetilde{r}(u,\tau) + 0.1\,\widetilde{\mathrm{RMSE}}_{\mathrm{inv}}(u,\tau) + 0.3\,\widetilde{N}_{\%}(u,\tau), \quad (5)$$

where each component is min–max scaled to $[0,1]$ within each sub-grid, and the RMSE component is inverted so that higher values indicate lower error. Higher scores indicate stronger relative performance among the mask–threshold configurations evaluated within the same sub-grid.

The weights are heuristic and reflect the objective of this study. Temporal correspondence receives the greatest weight because the analysis evaluates agreement in broad electricity-load trajectories, while retained temporal coverage is given substantial weight to avoid selecting configurations supported by a sparse subset of dates. RMSE receives a lower weight because the analysis prioritises temporal co-movement rather than direct estimation of disruption magnitude. The composite score is therefore a within-sub-grid ranking criterion, not an absolute or universally applicable measure of reliability.

**Table 2**
Pixel-scale observability diagnostics for Samar–Leyte (2013) evaluated across nested kernel supports (1×1, 3×3, 5×5). Metrics summarise how increasing spatial support modifies data availability, maximum dropout duration, and radiance variability.

| Kernel | Class | Spatial Completeness *SC* (%) | Temporal Completeness *TC* (%) | Max Dropout (days) |
|---|---|---|---|---|
| 1×1 | City centres | 31.1 | 31.1 | 14 |
| | Municipal centres | 33.5 | 33.5 | 13 |
| | Non-settlement Areas | 26.7 | 26.7 | 27 |
| 3×3 | City centres | 28.9 | 30.3 | 14 |
| | Municipal centres | 31.9 | 33.1 | 13 |
| | Non-settlement Areas | 24.9 | 26.4 | 26 |
| 5×5 | City centres | 26.5 | 28.9 | 14 |
| | Municipal centres | 30.2 | 32.7 | 13 |
| | Non-settlement areas | 24.2 | 26.2 | 27 |

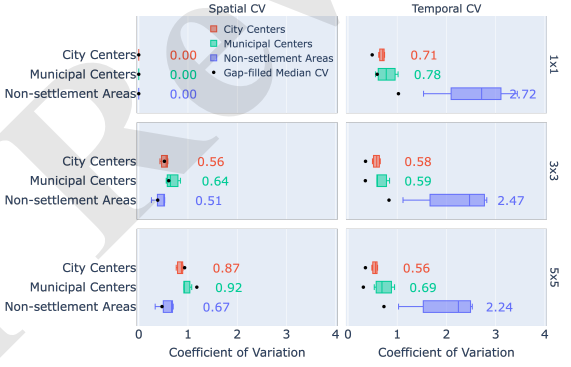


**Figure 4:** Temporal and spatial coefficients of variation (CV) of DNB-BRDF radiance across pixel kernel supports and settlement categories (2013). Median CV values for the gap-filled series are shown for comparison.

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**Table 3**
Settlement-scale observability diagnostics for Samar–Leyte (2013) stratified by GHSL Degree of Urbanization classes. Metrics characterise how data completeness and radiance stability vary across structurally distinct settlement regimes. Pixel share is the percentage of all analysed Samar–Leyte pixels assigned to each settlement class.

| Settlement Group | Pixel Share (%) | Observational Completeness | | | | | Signal Stability | |
|---|---|---|---|---|---|---|---|---|
| | (%) | Spatial (%) | Temporal (%) ≥25 | ≥50 | ≥75 | Max Dropout (Days) | Temporal CV | Spatial CV |
| Urban Centre | 0.55 | 36.6 | 47.4 | 35.9 | 24.7 | 40 | 1.56 | 1.16 |
| Urban Cluster | 14.39 | 38.6 | 50.7 | 37.9 | 24.2 | 45 | 1.23 | 1.42 |
| Non-urban Cells | 84.15 | 38.2 | 52.5 | 39.0 | 23.5 | 45 | 1.36 | 1.28 |
| Water | 0.99 | 66.6 | 57.8 | 47.7 | 38.8 | 26 | 7.49 | 1.41 |

## 3. Results

### 3.1. Local-Scale Observability and Stability

Table 2 summarises spatial completeness (SC), temporal completeness (TC), and maximum consecutive dropout duration for the three POI groups across the 1×1, 3×3, and 5×5 kernel supports. Across city and municipal centres, SC and TC generally range from approximately 27% to 35%, indicating that valid DNB-BRDF observations are available on roughly one-third of days. Table S2 resolves these group-level summaries by individual POI and shows greater variability among the land-based reference sites.

Increasing kernel support from 1×1 to 5×5 pixels produces only small changes in observation availability. Completeness generally changes by only a few percentage points across supports, while maximum consecutive dropout durations remain similar. Across the land-based POIs, maximum dropout periods are typically 11–18 days, showing that broader spatial support does not materially reduce prolonged observation gaps.

Radiance stability diagnostics (Fig. 4) show systematic differences by settlement context and kernel support. Temporal CV is lowest in city centres, intermediate in municipal centres, and highest in non-settlement locations, where median values exceed 2 across all supports. Spatial CV is effectively zero at the 1×1 support by definition and increases at 3×3 and 5×5 as additional pixels introduce within-kernel heterogeneity. Municipal centres show the highest spatial CV at the broader supports, followed by city centres, while non-settlement locations remain comparatively homogeneous spatially despite greater day-to-day variability.

A consistent product-level contrast is also evident in Fig. 4: gap-filled radiance shows lower temporal CV than directly observed DNB-BRDF radiance across settlement contexts and kernel supports. Regional and pixel-scale comparisons in Figs. S1–S2 show that this reduction in variability coincides with smoother gap-filled trajectories during periods of sparse direct observations, while Fig. S5 confirms the same pattern across representative urban, municipal, and non-settlement POIs. The pre–post Haiyan analysis in Fig. S4 further shows greater dispersion in ΔNTL at 1×1 support and more stable, but slightly smaller, change estimates at 5×5 support.

### 3.2. Settlement-Scale Observability Diagnostics

Table 3 reports settlement-scale observability diagnostics for Samar–Leyte using GHSL SMOD classes. Across the principal land-based settlement groups, spatial completeness is closely grouped, ranging from 36.6% in urban centres to 38.6% in urban clusters. Temporal completeness decreases substantially as the spatial validity requirement becomes stricter. Approximately 47–53% of days satisfy the ≥25% criterion, 36–39% satisfy the ≥50% criterion, and only 23–25% satisfy the ≥75% criterion. Thus, the reduction in admissible days with increasing completeness thresholds is observed across urban and non-urban classes rather than within a single settlement regime.

Maximum dropout duration is similarly prolonged across the grouped settlement classes. Urban centres show a maximum dropout of 40 days, while urban clusters and non-urban cells reach 45 days. Table S3 disaggregates these grouped results into the constituent GHSL SMOD classes, showing that subclass-level dropout durations range from 33 to 45 days. The supplementary results also reveal greater variation in radiance stability than is apparent from the grouped values, particularly between dense urban, peri-urban, and rural subclasses.

Temporal CV varies more strongly across GHSL SMOD classes than spatial CV. Among the land-based classes, temporal CV ranges from approximately 0.8 to 2.2, with the highest variability occurring in rural clusters. Spatial CV shows a narrower range, with higher values in spatially mixed urban and rural classes than in very low-density rural cells.

Settlement composition is strongly imbalanced across Samar–Leyte. Urban centres account for only 0.55% of the analysed pixels and urban clusters for 14.39%, whereas non-urban cells account for 84.15%. Figure S6 shows the spatial structure of the GHSL thematic masks derived from these classes. Urban-focused masks are compact and concentrated around contiguous high-radiance settlements, while progressively broader masks incorporate dispersed peri-urban, rural, and low-emission pixels.

The global temporal completeness map in Fig. 5 places the Samar–Leyte results within a wider geographical pattern. Annual completeness is generally higher across many arid and temperate regions and lower across equatorial and

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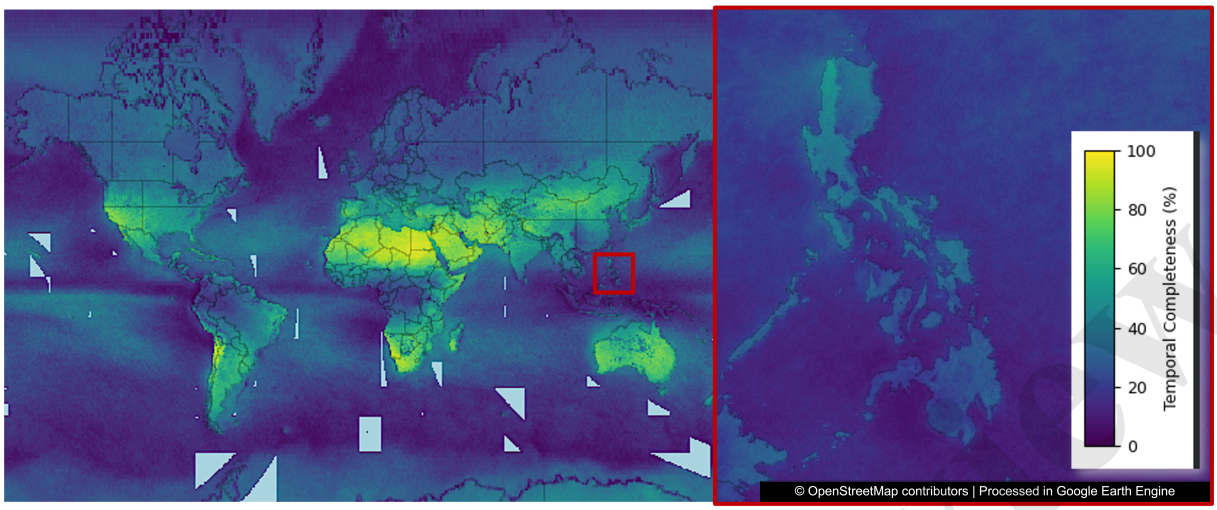


**Figure 5:** Global temporal completeness of daily VIIRS Black Marble DNB-BRDF observations in 2013, with the Philippines shown in detail. Lower completeness is concentrated across equatorial and monsoonal regions, including maritime Southeast Asia. White areas indicate locations for which no temporal-completeness value was available. VNP46A2 data were processed in Google Earth Engine; inset basemap © OpenStreetMap contributors.

monsoonal regions. The Philippines forms part of a coherent low-completeness belt across maritime Southeast Asia. The long-term Samar–Leyte diagnostics in Fig. S3 extend this comparison beyond 2013 and show that reduced spatial completeness recurs throughout the 2012–2024 record, with pronounced seasonal and interannual variation rather than being confined to the Haiyan period.

### 3.3. Reliability-Qualified Alignment Between Stratified NTL and Sub-Grid Load

Figure 6 summarises the systematic sweep of spatial completeness thresholds ($\tau$) across cumulative GHSL thematic masks. Colour saturation at the reference values highlights broader regions of favourable association, error, and temporal coverage, while bold values identify the top-performing configurations within each metric. The composite score is used separately to rank complete mask–threshold configurations. The GHSL thematic masks represent progressively broader settlement supports, from dense urban classes to mixed and rural-dominant classes, with mask examples shown in Fig. S6.

Alignment varies systematically with both cumulative GHSL thematic mask and spatial completeness threshold. Urban-focused masks produce the strongest correspondence with NGCP sub-grid electricity load. Within the G1–G3 masks, Pearson $r$ generally exceeds 0.80 across a broad low- to mid-threshold range, reaching approximately 0.88, while minimum RMSE values approach 0.50. Broader rural-dominant masks show lower correlations and higher RMSE across most thresholds.

Retained temporal coverage declines continuously as $\tau$ increases. At low thresholds, approximately half of the common analysis period remains available for several masks, whereas the strictest thresholds retain fewer than 10% of dates. The metric surfaces therefore do not identify a single threshold that simultaneously maximises correlation, minimises error, and preserves temporal coverage. Instead, they show a broad high-alignment region for urban-focused masks combined with a progressive coverage penalty as spatial completeness requirements become stricter.

Figure 7 presents the Samar–Leyte alignment at $\tau$=10%, the configuration associated with the highest composite score. Urban-focused masks follow the principal temporal structure of NGCP electricity load more closely than broader rural-dominant masks, including the decline following Typhoon Haiyan and the subsequent recovery trajectory.

Supplementary threshold diagnostics provide additional examples of this pattern. Figure S7 shows that increasing $\tau$ changes both the continuity and apparent amplitude of the Haiyan-period radiance trajectory. Figure S8 compares settlement-stratified DNB-BRDF radiance and NGCP load at $\tau$ = 10%, 50%, and 90%, illustrating the progressive loss of temporal coverage as the spatial-support requirement increases. At $\tau$ = 10%, more observations are retained but the NTL series shows greater high-frequency variability. At $\tau$ = 50%, the broad temporal trajectory remains visible, but fewer dates are retained. At $\tau$ = 90%, the series becomes highly fragmented and substantial portions of the temporal record are unavailable.

### 3.4. Transferability of Reliability-Qualified Configurations

Table 4 reports the highest-scoring configuration for each Visayas sub-grid. Urban-focused masks are selected in all five regions, with G1 or G2 providing the strongest overall performance. The selected spatial completeness threshold is 10% for Samar–Leyte, Negros, and Panay, 20% for Bohol, and 60% for Cebu. Correlations range from $r$ = 0.78 to



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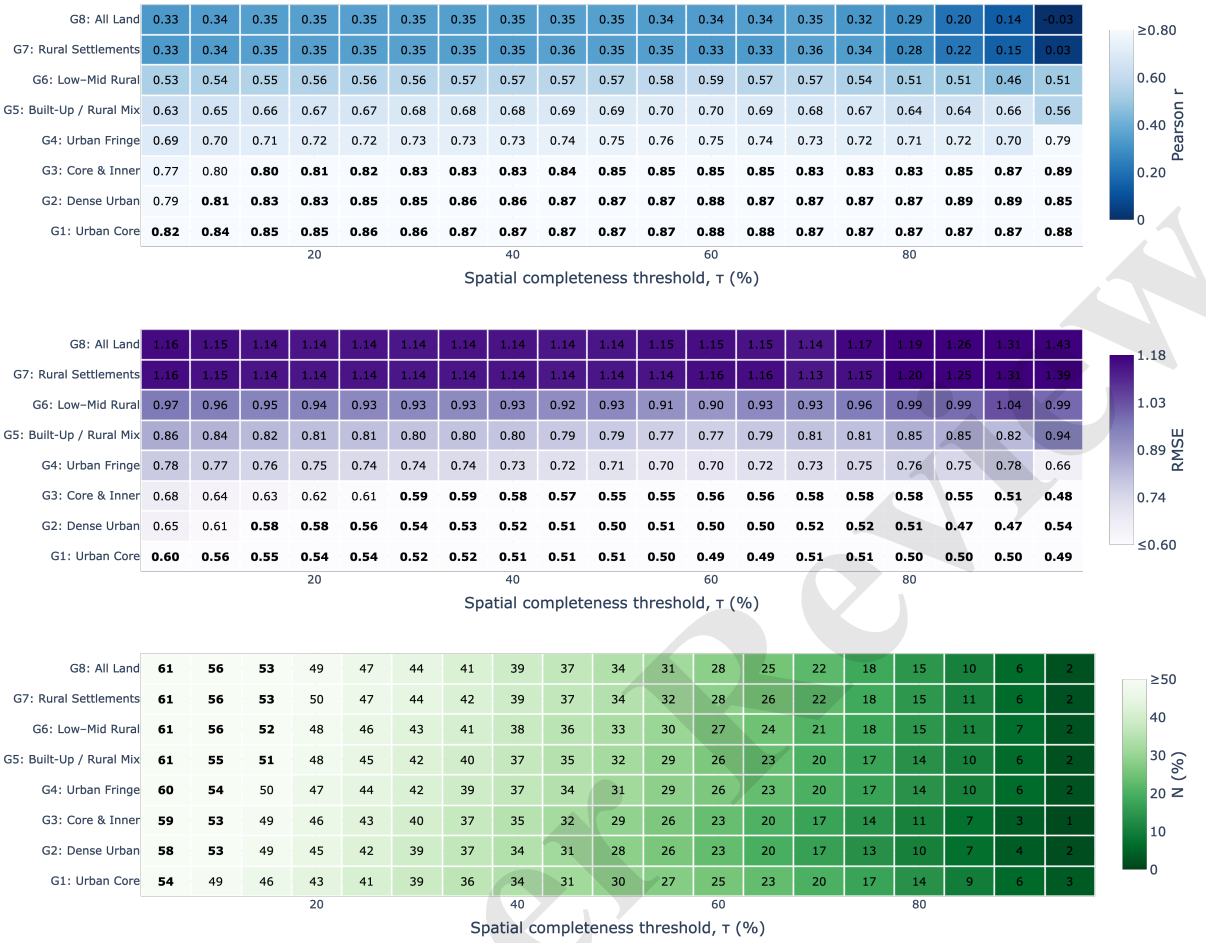


**Figure 6:** Systematic sweep of spatial completeness thresholds ($\tau$) across cumulative GHSL thematic masks (G1–G8). Cell values report Pearson $r$, RMSE, and retained temporal coverage $N_{\%}$. Colour scales saturate at the interpretive reference values $r = 0.80$, RMSE $= 0.60$, and $N_{\%} = 50\%$ to emphasise broader regions of favourable performance; these values are visual guides rather than formal acceptance thresholds. Bold values identify the top-performing configurations within each metric.

**Table 4**
Highest-scoring reliability-qualified configurations for each Visayas sub-grid (2012–2024).

| Sub-grid | Thematic Mask (GHSL range) | $\tau$ (%) | Composite | Pearson $r$ | RMSE | Temporal Coverage (%) |
|---|---|---|---|---|---|---|
| Samar–Leyte | G2: Dense Urban (23–30) | 10 | 0.93 | 0.84 | 0.57 | 49.81 |
| Bohol | G1: Urban Core (30-only) | 20 | 0.88 | 0.86 | 0.54 | 28.71 |
| Cebu | G1: Urban Core (30-only) | 60 | 0.74 | 0.78 | 0.67 | 21.38 |
| Negros | G1: Urban Core (30-only) | 10 | 0.90 | 0.78 | 0.66 | 38.69 |
| Panay | G2: Dense Urban (23–30) | 10 | 0.89 | 0.80 | 0.64 | 43.89 |

0.86, while retained temporal coverage ranges from approximately 21% to 50%. Because the composite components are normalised independently within each sub-grid, the scores rank configurations locally and are not interpreted as direct measures of relative performance between sub-grids.

All five selected configurations use urban-focused thematic masks. Samar–Leyte and Panay select the G2 Dense Urban mask, while Bohol, Cebu, and Negros select the G1 Urban Core mask. The selected spatial completeness thresholds vary substantially among sub-grids. Samar–Leyte, Negros, and Panay achieve their highest composite scores at $\tau = 10\%$, Bohol at $\tau = 20\%$, and Cebu at $\tau = 60\%$.

The selected configurations produce correlations of $r = 0.78$–$0.86$ and RMSE values of 0.54–0.67. Temporal coverage ranges from approximately 21% in Cebu to 50% in Samar–Leyte. Thus, four of the five sub-grids achieve their

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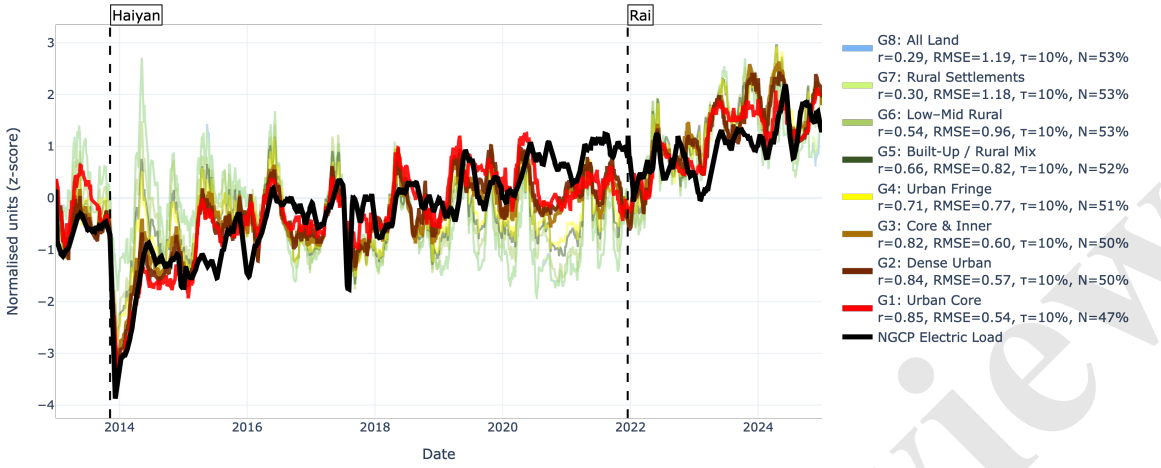


**Figure 7:** Alignment between GHSL-stratified, reliability-qualified DNB-BRDF radiance and NGCP sub-grid electricity load for Samar–Leyte at $\tau = 10\%$, the threshold associated with the highest composite score. Series are z-score normalised and smoothed using a 30-day centred moving average.

highest composite scores under relatively permissive thresholds that preserve approximately 28–50% of the daily record. Cebu differs, with its highest-ranked configuration occurring under a stricter threshold and retaining approximately 21% of days.

The five highest-ranked configurations for each sub-grid are reported in Table S10. These expanded rankings show that several mask–threshold combinations often produce similar composite scores within the same sub-grid. Despite variation in the selected threshold, the highest-ranked configurations remain concentrated in the G1–G3 urban-focused masks.

Figure 8 shows the time-domain alignment for Bohol, Cebu, Negros, and Panay using the highest composite-score configuration identified for each sub-grid. Across the four additional sub-grids, the reliability-qualified urban NTL series reproduce the principal temporal structure of NGCP load. Major disruption periods, including Typhoon Haiyan and Typhoon Rai, remain visible, although the amplitude and temporal definition of the NTL response vary among sub-grids.

## 4. Discussion

### 4.1. Observability as a Structural Constraint on Daily NTL Interpretation

The results show that daily VIIRS Black Marble observations in Samar–Leyte are highly intermittent across both pixel and settlement scales. At the POI scale, valid DNB-BRDF observations are available on fewer than one-third of days for most sites, and increasing support from 1×1 to 5×5 pixels produces only minor changes in completeness. At the settlement scale, temporal completeness also declines sharply as admissibility thresholds become stricter. These findings indicate that the limiting factor is not simply pixel-level noise, but repeated absence of valid observations over large areas.

This distinction is central to the reliability argument. Spatial aggregation can reduce radiance variability on retained days, but it does not recover days lost to cloud obstruction. The long-term completeness diagnostics in Fig. S3 reinforce this point by showing that incomplete sampling is embedded within seasonal and interannual completeness structure across the 2012–2024 record. The dropout values reported in Tables S2–S3 further show that completeness percentages alone are insufficient: two locations can have similar annual completeness but very different maximum gap lengths. For disaster applications, the duration of observation gaps matters because extended dropout windows can overlap with disruption and early recovery periods.

The contrast between DNB-BRDF and gap-filled radiance further clarifies the difference between continuity and observability. Gap-filled data reduce temporal CV and produce smoother trajectories, but the supplementary diagnostics show that this smoothness is introduced through temporal propagation rather than new observations. Figures S1, S2, and S5 show that the gap-filled series can dampen abrupt radiance changes around Typhoon Haiyan, particularly at fine spatial supports. This supports the use of DNB-BRDF as the primary signal for reliability-sensitive disaster analysis, with gap-filled radiance retained as a comparator rather than treated as equivalent observational evidence.

These findings address a core assumption in daily NTL studies: that near-daily products provide near-continuous information. In cloud-impacted tropical regions, the nominal daily temporal resolution of the product does not translate into daily surface observability. The relevant question is therefore not only whether a daily NTL value exists, but



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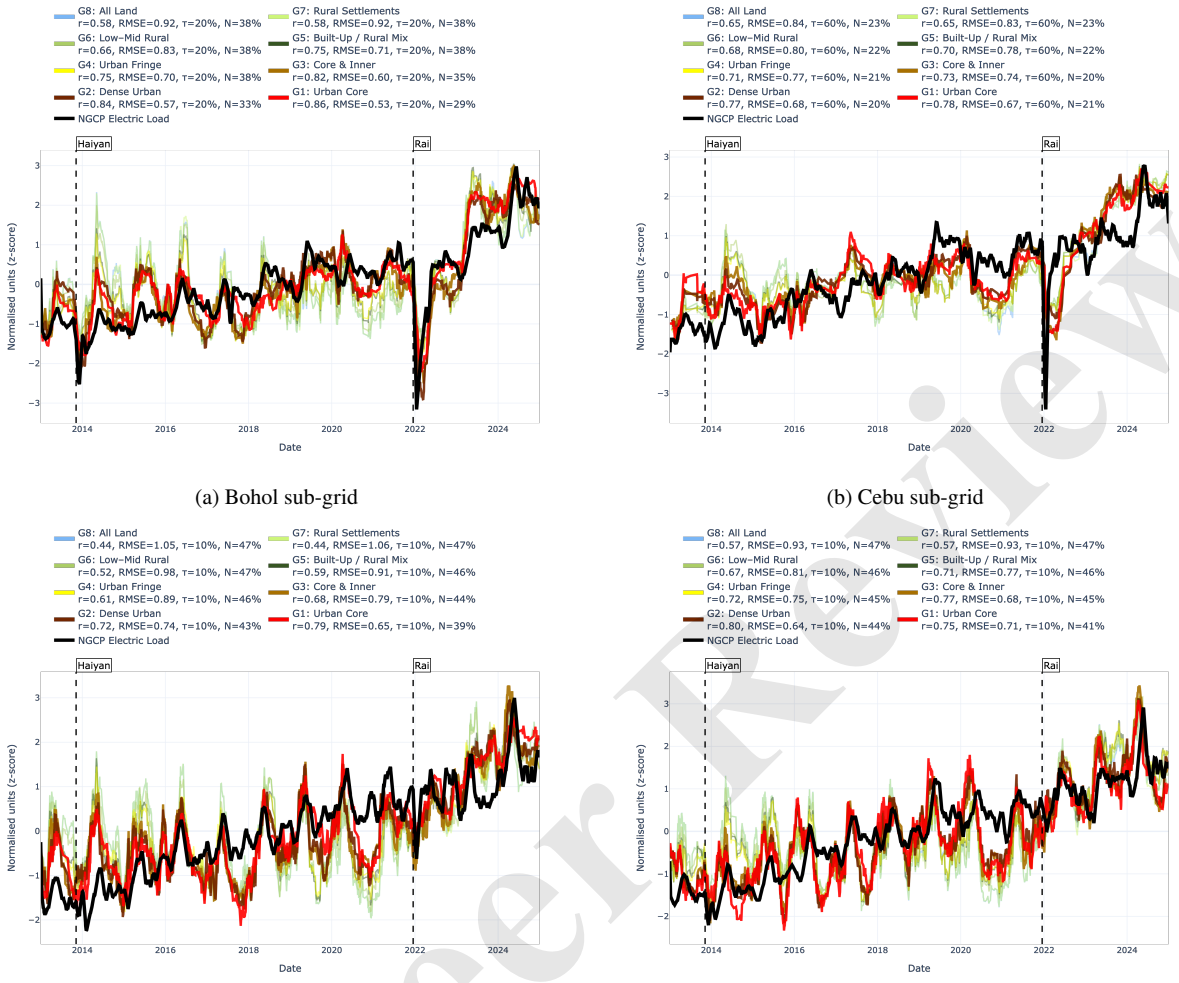


(a) Bohol sub-grid

(b) Cebu sub-grid

(c) Negros sub-grid

(d) Panay sub-grid

**Figure 8:** Reliability-qualified NTL–NGCP alignment for the Bohol, Cebu, Negros, and Panay sub-grids using the highest composite-score configuration identified for each area. Coloured curves represent GHSL-stratified DNB-BRDF radiance, while the black curve represents NGCP sub-grid electricity load.

whether that value is supported by sufficient direct observations to justify interpretation.

## 4.2. Settlement Structure Conditions the Reliability of NTL Signals

Settlement stratification substantially changes how the NTL signal behaves. Urban centres and clusters occupy a small fraction of the Samar–Leyte domain, but they contain the strongest and most persistent artificial lighting signal. Non-urban classes dominate the spatial area and contribute lower-intensity radiance whose variability differs substantially among rural subclasses. Without settlement stratification, regional aggregation would be weighted toward land-cover contexts that are less directly representative of electricity-dependent activity.

The settlement-scale results show that observability is not reducible to completeness alone. Urban and rural classes may show similar spatial completeness values, but their radiance stability, baseline brightness, and spatial morphology differ. Figure S6 illustrates why this matters: urban masks are compact and concentrated around high-emission areas, whereas broader rural and non-urban masks are spatially dispersed and more heterogeneous. This morphology affects both signal stability and the distribution of valid observations within each aggregation unit.

The alignment results are consistent with this settlement structure. Urban-dominant classes produce the strongest NTL–load correspondence, while broader masks dilute the signal by incorporating lower-emission and more heterogeneous pixels. This does not mean that rural areas are unimportant for disaster recovery analysis. Rather, it shows that the direct radiance–load relationship is strongest where lighting is dense, persistent, and more closely coupled to

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grid-connected infrastructure. For rural and low-density areas, NTL may still contain useful information, but its interpretation requires additional caution and likely additional contextual data.

### 4.3. Completeness Thresholds Define a Context-Dependent Trade-Off

The spatial completeness threshold is not merely a technical filtering choice. It changes both the observational support and temporal representation of the NTL series. Increasing $\tau$ excludes days with limited valid-pixel coverage, but it also removes progressively larger portions of the daily record. Consequently, the threshold associated with the strongest correlation or lowest RMSE does not necessarily produce the highest composite score once temporal coverage is considered.

The Samar–Leyte heatmaps show that urban-focused masks maintain high correlation and low RMSE across a broad low- to mid-threshold range rather than at a single sharply defined optimum. The supplementary threshold comparisons further show that stricter thresholds alter the apparent continuity and amplitude of the disaster-period signal. At $\tau$=90%, large portions of the temporal trajectory are unavailable, while lower thresholds preserve more of the sequence but include days supported by fewer directly observed pixels.

The cross-sub-grid results demonstrate that this trade-off varies geographically. Samar–Leyte, Negros, and Panay achieve their highest composite scores at $\tau$=10%, Bohol at $\tau$=20%, and Cebu at $\tau$=60%. Reliability qualification should therefore not impose a universal completeness threshold. Instead, the admissibility setting should be selected through an explicit sensitivity analysis that jointly considers settlement support, association strength, error, and retained temporal coverage.

### 4.4. Conditional Transferability Across Tropical Island Sub-Grids

The transferability analysis shows that the most consistent element of the reliability framework is settlement stratification rather than a single spatial completeness threshold. Urban-focused masks are selected across all five sub-grids, indicating that compact, persistently illuminated settlement classes provide the most stable basis for alignment with sub-grid electricity load. In contrast, the highest-scoring threshold varies from 10% to 60%, demonstrating that no universal admissibility threshold applies across the Visayas.

The variation in selected thresholds reflects the trade-off embedded in the composite score. Lower thresholds preserve more of the temporal record but admit days with weaker spatial support, whereas stricter thresholds increase observational support at the cost of temporal coverage. Under the adopted composite-score formulation, the $\tau = 10\%$ configurations receive the highest overall ranking in Samar–Leyte, Negros, and Panay. Bohol shows a similar balance at $\tau = 20\%$. Cebu differs because its highest composite score occurs at $\tau = 60\%$, although neighbouring G1 thresholds produce similar scores and retain somewhat greater temporal coverage.

The expanded rankings in Table S10 further show that the performance surface is not characterised by a single sharply defined optimum. Several configurations within each sub-grid have similar composite scores, but they remain concentrated in urban-focused masks. The transferable result is therefore not a fixed threshold, but the repeated selection of urban-focused masks and a systematic procedure for balancing alignment performance against temporal coverage.

These findings refine the conditions under which daily NTL can be used across tropical island systems. Spatial filtering can be generalised around urban-focused settlement masks, whereas the completeness threshold should be determined for each sub-grid through an explicit sensitivity sweep. Transferability therefore lies in the diagnostic framework and selection logic, not in imposing one parameter setting across all locations.

### 4.5. Implications for Disaster Monitoring with Daily NTL

These results refine how daily NTL should be used for disaster impact and recovery analysis in cloud-prone regions. The contribution is not simply that NTL can align with electricity load. Prior work has already demonstrated the value of Black Marble for outage and restoration monitoring in settings such as Puerto Rico (Román et al., 2019). The contribution here is to define the conditions under which such interpretation becomes defensible in one of the most observationally difficult environments for daily optical monitoring.

The framework shifts the analytical question from "Can daily NTL detect disruption?" to "Under what observational conditions can daily NTL be interpreted as evidence of disruption or recovery?" This is especially important in tropical regions where disaster exposure is high but valid observations are sparse. Treating observability as a prerequisite changes the role of preprocessing: observational support, settlement structure, and retained temporal coverage become part of the evidence base rather than supplementary diagnostics.

The results also show why location- and disaster-agnostic use of NTL requires reliability qualification. A universal NTL recovery metric cannot be assumed if the underlying observation record is intermittent, gap-filled, or unevenly distributed across settlement types. However, the framework itself is transferable because it is based on diagnostic conditions rather than event-specific calibration. The framework is designed for extension to other hazards, regions, and infrastructure contexts, provided that observability is evaluated before impact or recovery metrics are interpreted.

Overall, the study pushes the usable limits of daily VIIRS NTL by showing that even in cloud-impacted tropical environments, the signal is not unusable. It is conditionally usable. When direct observations are separated from gap-filled continuity, when settlement structure is used to isolate stable lighting contexts, and when completeness thresholds

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are explicitly enforced, daily NTL can provide a defensible proxy for broad sub-grid electricity dynamics. Without these conditions, the same data risk being interpreted as continuous evidence when they are, in practice, intermittent observations shaped by cloud cover, settlement morphology, and preprocessing choices.

## 5. Conclusion

Daily NASA Black Marble observations in humid tropical regions are frequently interrupted by cloud-driven observation gaps and cannot be assumed to form continuous daily records. In Samar–Leyte and across neighbouring Visayas sub-grids, these gaps substantially restrict temporal coverage. Nevertheless, directly observed DNB-BRDF radiance shows strong correspondence with sub-grid electricity load when settlement structure and spatial completeness are explicitly accounted for.

The principal contribution of this study is the formalisation of when daily NTL interpretation is observationally defensible. Spatial completeness, temporal completeness, maximum dropout duration, and radiance stability are treated as primary diagnostic results rather than secondary quality-control information. Under the adopted weighting scheme, urban-focused masks consistently provide the strongest alignment with electricity load, while the threshold associated with the highest composite score varies among sub-grids from 10% to 60%. The transferable finding is therefore a reliability-qualification procedure and a consistent dependence on urban settlement structure, not a universal completeness threshold.

By placing observability diagnostics before impact inference, the framework provides a reproducible basis for evaluating daily NTL in persistently cloudy environments. Future work should develop impact and recovery metrics specifically designed for intermittently observed records and test the framework across additional cloud regimes, hazards, and infrastructure systems.

## Data Availability Statement

Processed data and analysis outputs supporting this study are available on Zenodo at https://doi.org/10.5281/zenodo.21051901. The reproducible code, including Google Earth Engine scripts, Python workflows, notebooks, and documentation, is archived with the Zenodo release. NASA Black Marble VNP46A2 and GHSL-SMOD source data are available from their official repositories and are not redistributed. Hourly electricity-demand data were obtained from the publicly accessible National Grid Corporation of the Philippines Operations portal; original NGCP files are not redistributed, but derived analysis-ready variables and processing documentation are provided subject to source reuse conditions.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## CRediT authorship contribution statement

**Rene Jr. L. Principe:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Visualization, Writing - Original draft, Writing - Review & Editing. **Karin Reinke:** Supervision, Formal analysis, Investigation, Writing - Review & Editing, Project administration. **Simon Jones:** Supervision, Formal analysis, Writing - Review & Editing, Project administration.

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